

HW2. Start early. Otherwise, it will be difficult to finish on time

FOR LOOP



Parallel (broadcasting) numpy



Gaussian Mixture Model

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Outline

- Overview 
- Gaussian Mixture Model
- The Expectation-Maximization Algorithm

Recap

Conditional probabilities:

$$p(A, B) = p(A|B)p(B) = p(B|A)p(A)$$

Bayes rule:

$$p(A|B) = \frac{p(A, B)}{p(B)} = \frac{p(B|A)p(A)}{p(B)}$$

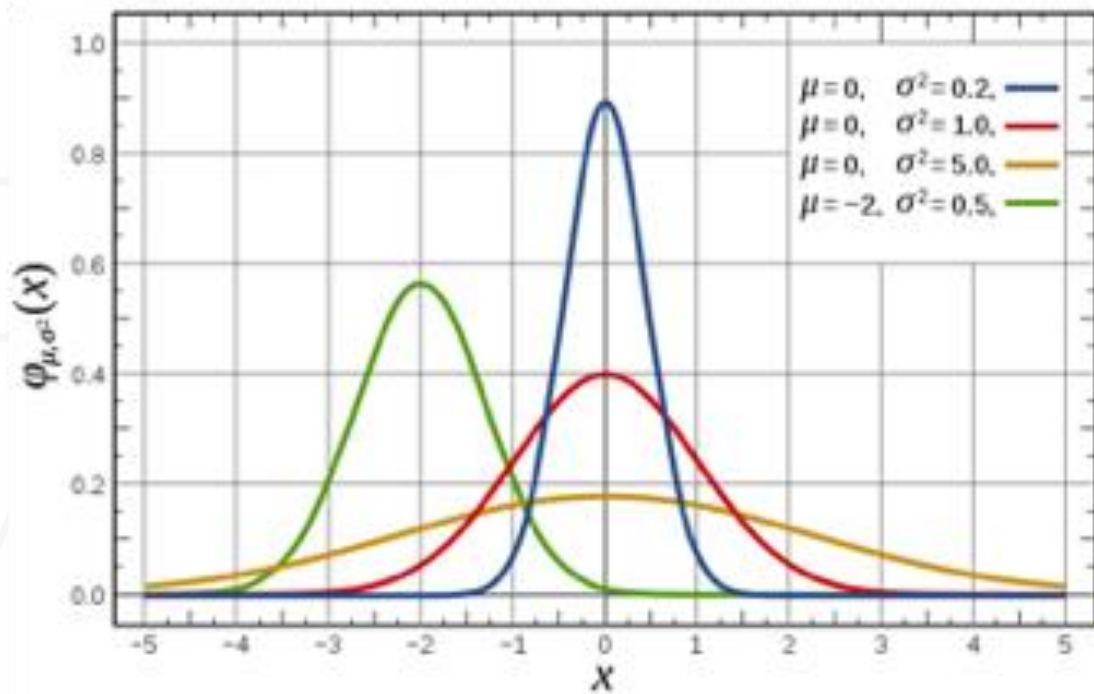
$$p(A = 1) = \sum_{i=1}^K p(A = 1, B_i) = \sum_{i=1}^K p(A|B_i) p(B_i)$$

	Tomorrow=Rainy	Tomorrow=Cold	P(Today)
Today=Rainy	4/9	2/9	$[4/9 + 2/9] = 2/3$
Today=Cold	2/9	1/9	$[2/9 + 1/9] = 1/3$
P(Tomorrow)	$[4/9 + 2/9] = 2/3$	$[2/9 + 1/9] = 1/3$	

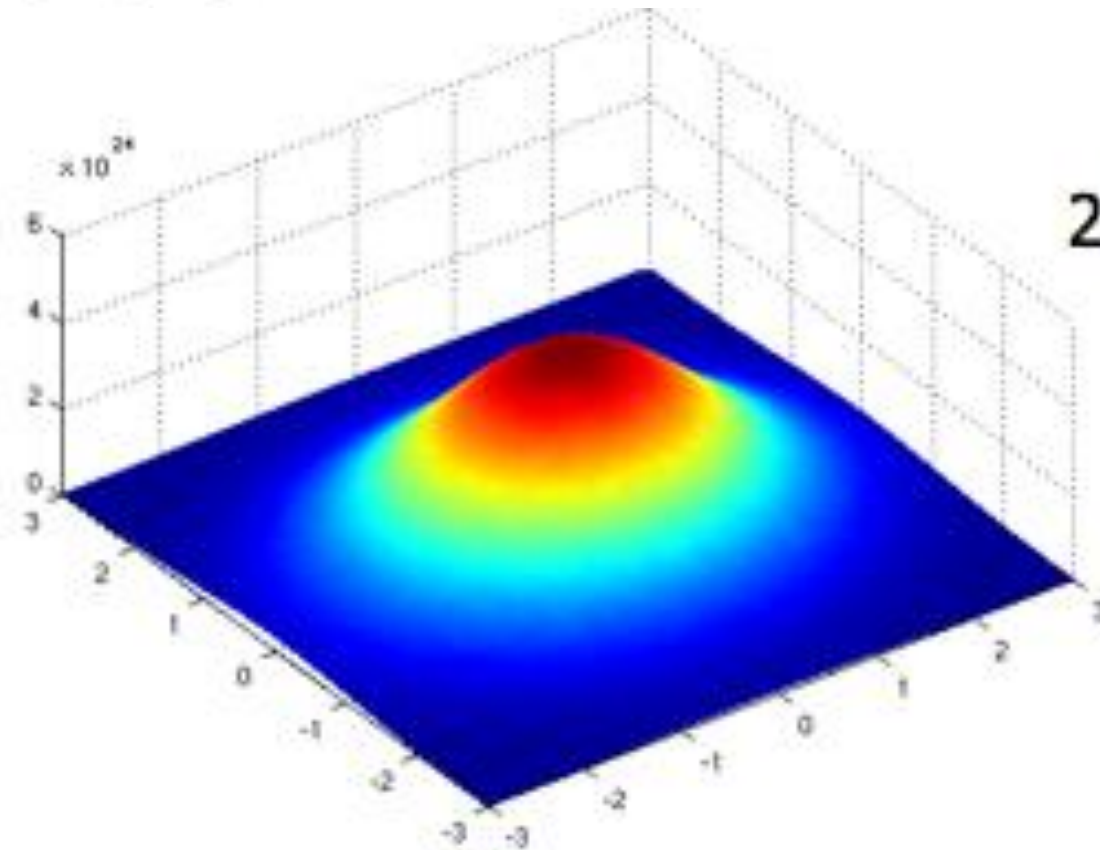
$P(\text{Tomorrow} = \text{Rainy}) =$

Gaussian Distribution

1-d Gaussian



$$N(\mu, \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x^{\{1\}} - \mu)^2}{2\sigma^2}}$$



2-d Gaussian

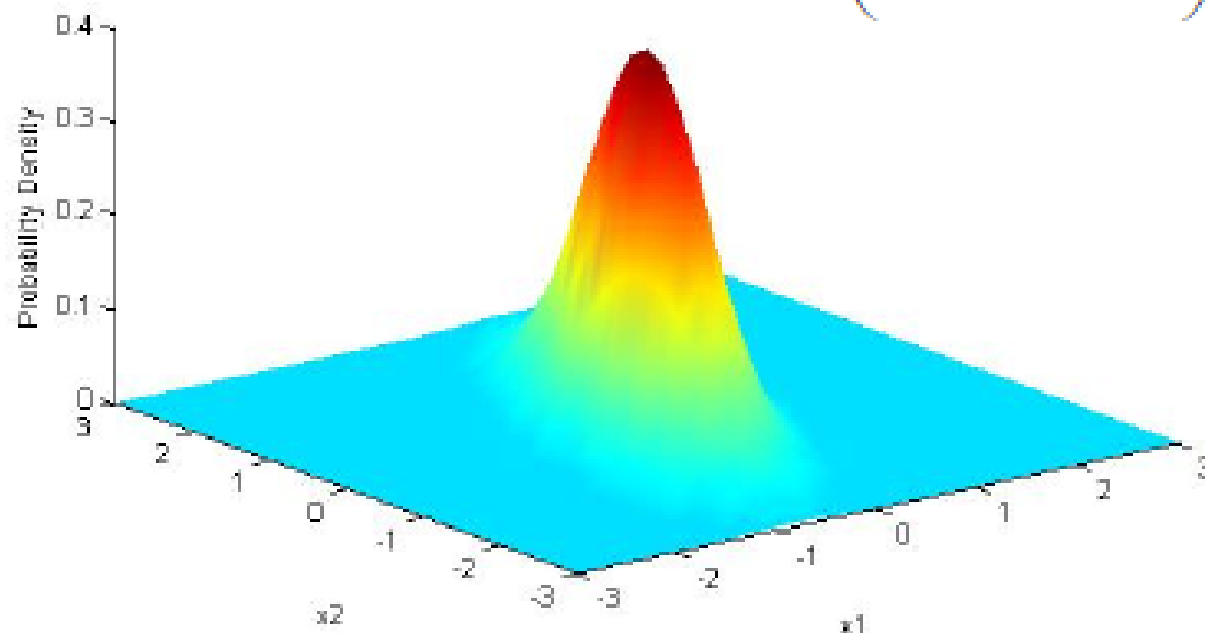
What is a Gaussian?

For d dimensions, the Gaussian distribution of a vector $x^{\{\cdot\}} = (x_1, x_2, x_3, \dots, x_d)^T$ is defined by:

$$N(x | \mu, \Sigma) = \frac{1}{(2\pi)^{d/2} \sqrt{|\Sigma|}} \exp\left(-\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu)\right)$$

where μ is the mean and Σ is the covariance matrix of the Gaussian.

Example: $\mu = (0,0)^T$ $\Sigma = \begin{pmatrix} 0.25 & 0.30 \\ 0.30 & 1.00 \end{pmatrix}$



Is Gaussian Concave?

$$\begin{aligned} N(x|\mu, \sigma) &= N(\mu, \sigma) = p(x) = f(x) = p(x|(\mu, \sigma)) = f(x|(\mu, \sigma)) \\ &= p(x|\theta) = f(x|\theta) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x^{\{1\}} - \mu)^2}{2\sigma^2}} \end{aligned}$$

Gaussian Optimization: How does being log concave help?

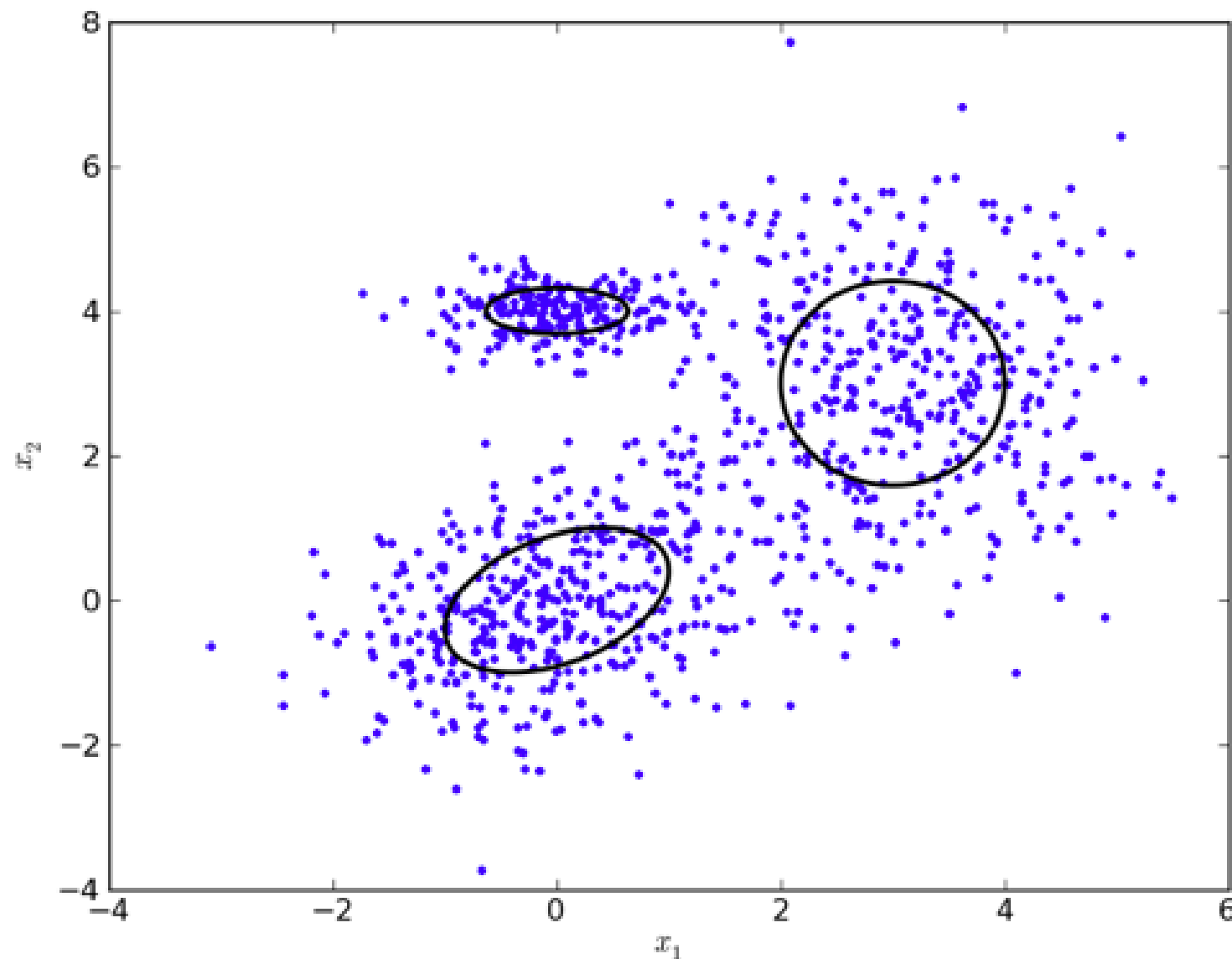
Log of sum of gaussians vs. sum of log of gaussians

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Hard Clustering Can Be Difficult

- Hard Clustering: K-Means, Hierarchical Clustering, DBSCAN



Towards Soft Clustering

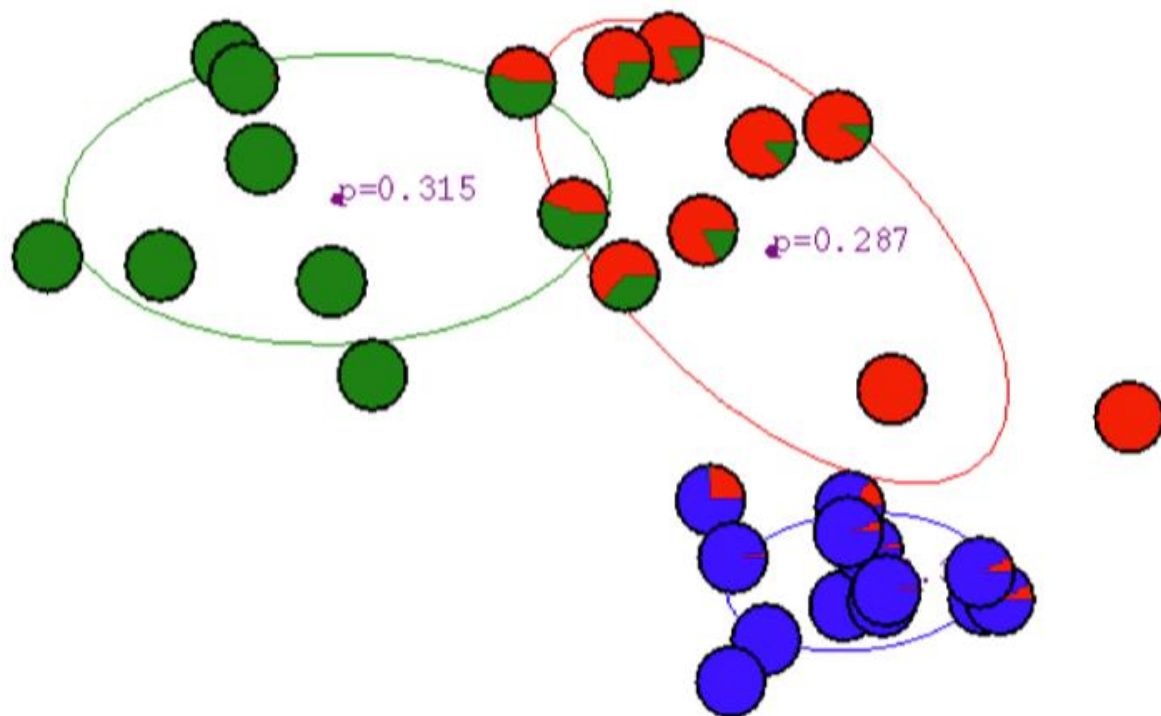
- **K-means**

- hard assignment:** each object belongs to only one cluster

$$\theta_i \in \{\theta_1, \dots, \theta_K\}$$

- **Mixture modeling**

- soft assignment:** probability that an object belongs to a cluster



Why Soft Clustering?

- **Captures Uncertainty:**

Points near cluster boundaries are assigned probabilities, not forced into one group. Equal probabilities suggest boundary points.

- **Provides Probabilistic Insight:**

Gives likelihood of belonging to each cluster → helpful for decisions that depend on confidence.

- **Supports Better Decision-Making:** E-commerce: Target customers based on % likelihood of being premium vs. budget shoppers.

- **Hard Clustering (k-means):** Customer A is assigned to “Budget Shopper” cluster. Customer B is assigned to “Premium Shopper” cluster — done

- **Soft Clustering (GMM):**

- Customer A: 70% Budget Shopper, 30% Occasional Premium Buyer

- Customer B: 50% Budget, 50% Premium

- Now you can Target Customer A with budget-friendly items most of the time but occasionally show premium items. Treat Customer B as a swing buyer — push offers to see which direction they lean over time.

Why Soft Clustering?

- **Supports Better Decision-Making:**

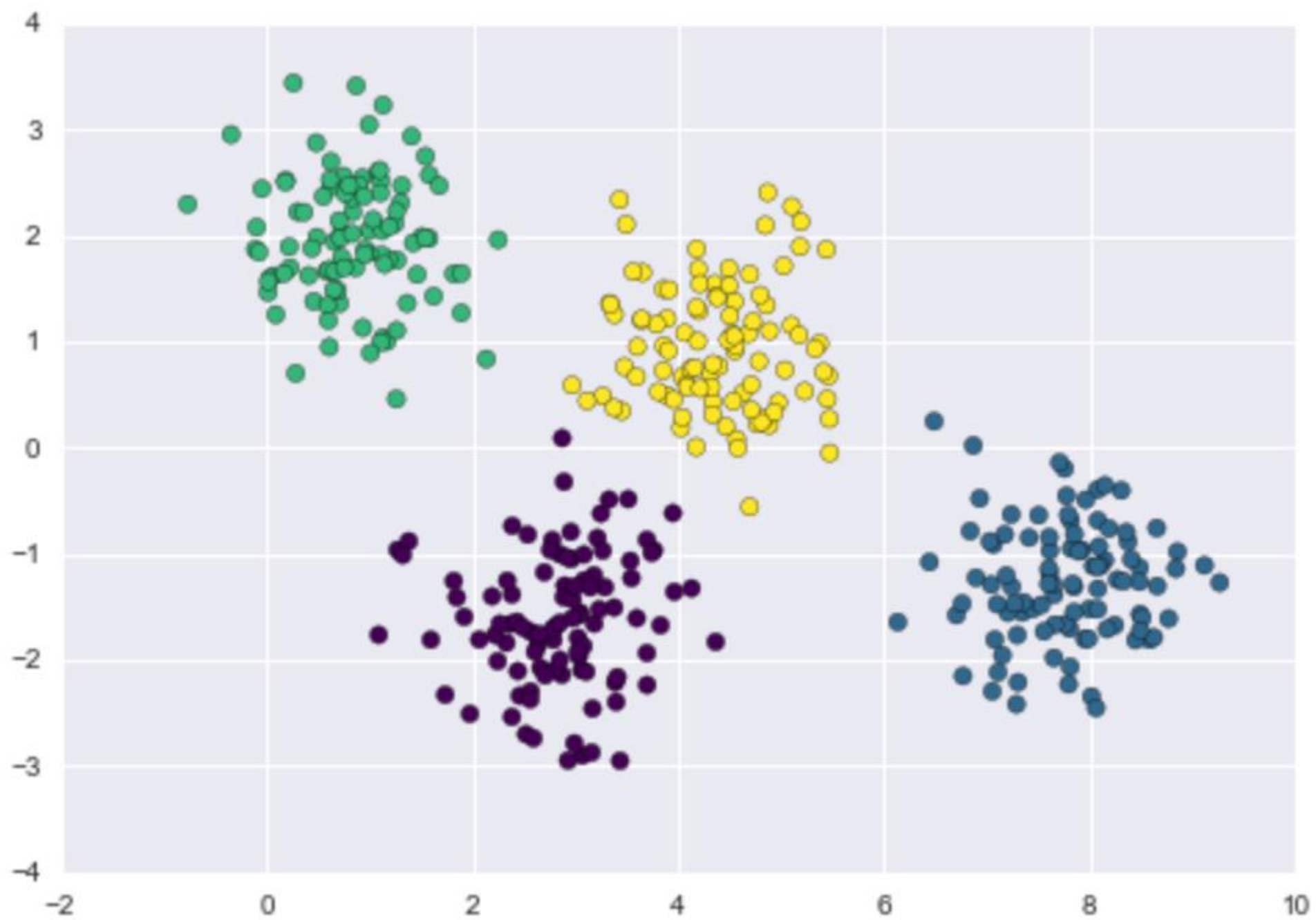
- Finance: Flag transactions that are *somewhat* suspicious instead of just yes/no.

- **Enables Anomaly Detection:**

Outliers have very low probability under all clusters → easy to detect.

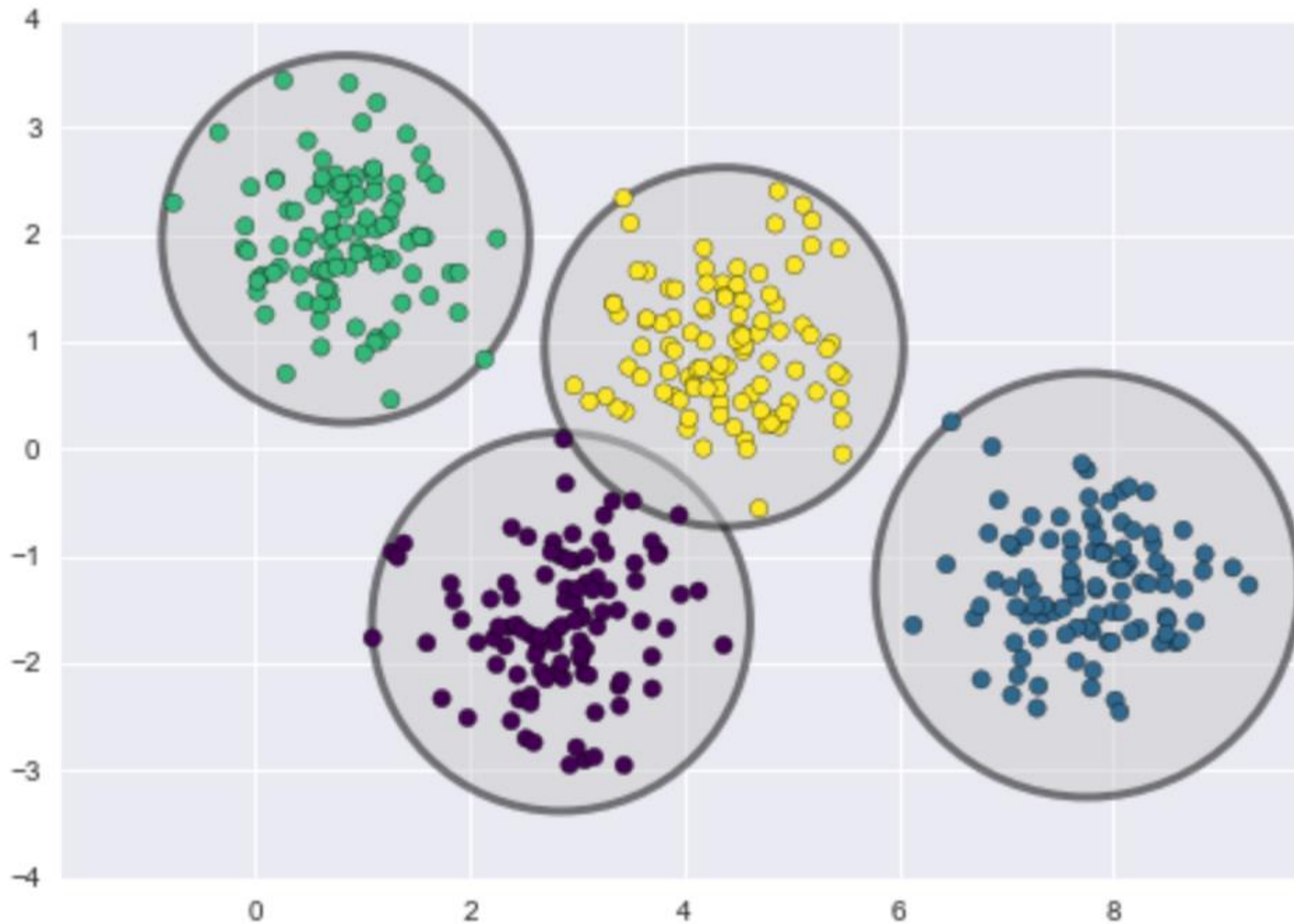
- **Models Different Cluster Shapes:**

Learns ellipses instead of just spheres — handles elongated or uneven clusters.



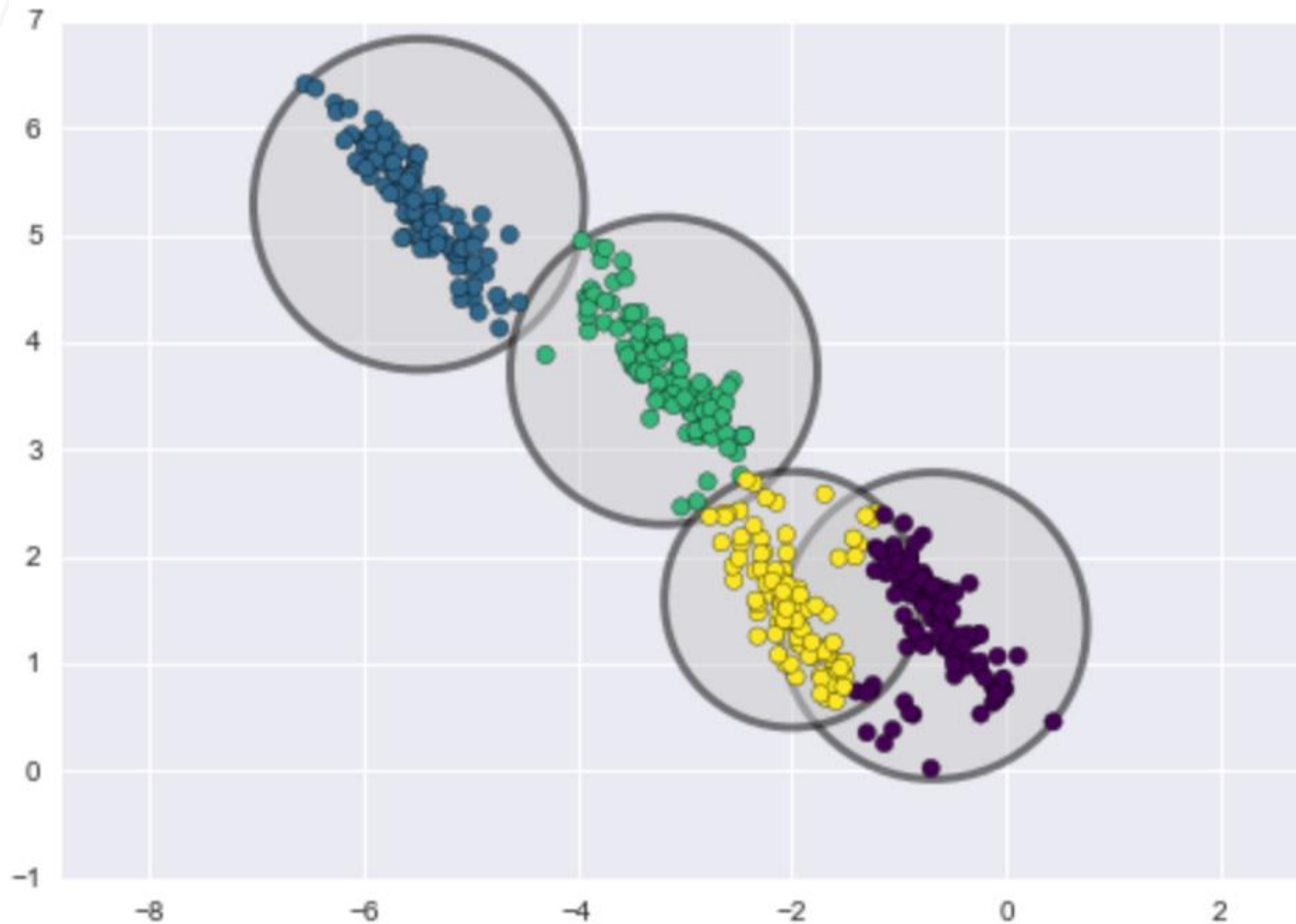
This is an excerpt from the [Python Data Science Handbook](#) by Jake VanderPlas

Let's run K-Means on the dataset

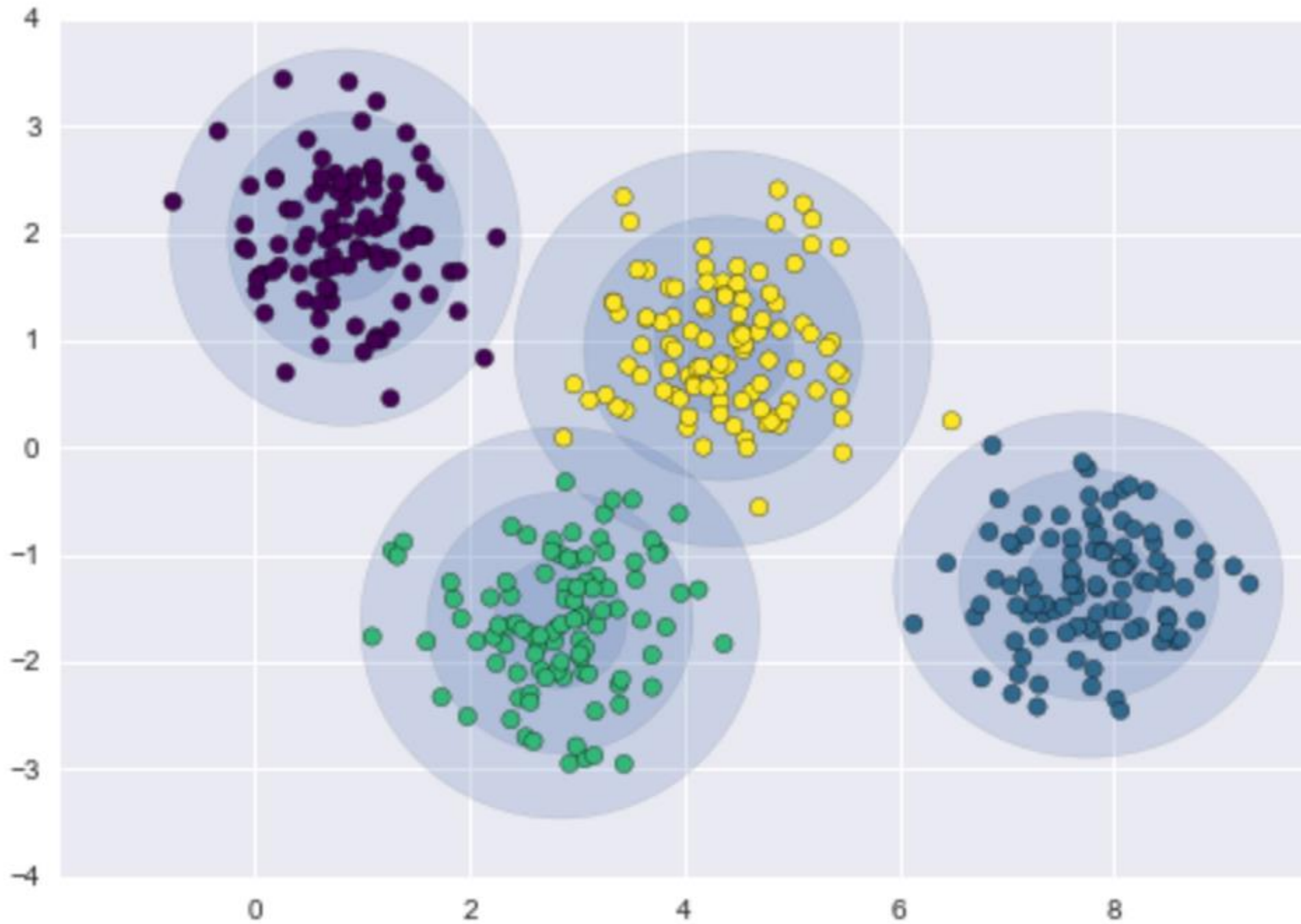


How many parameters to optimize in kmeans?

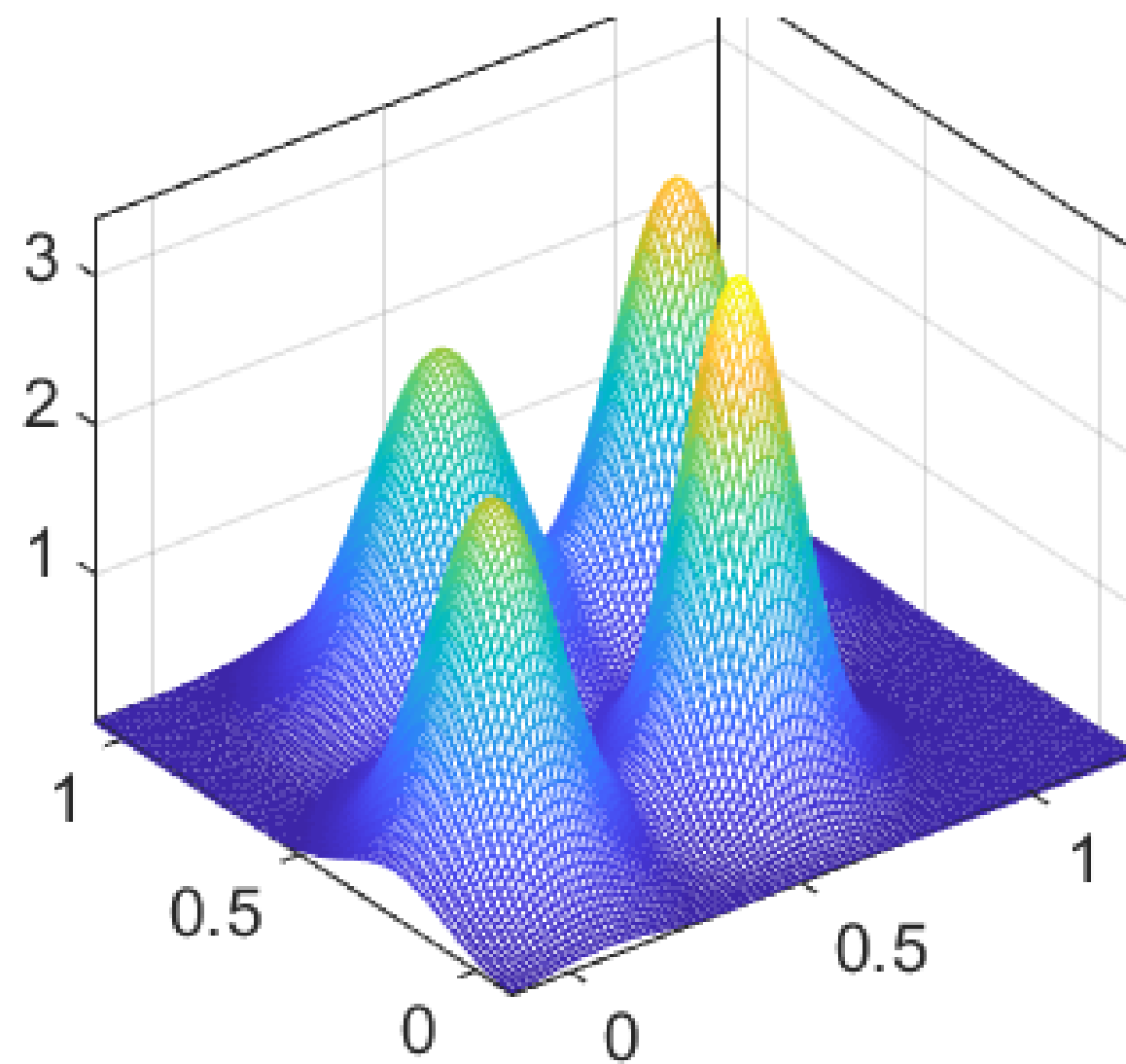
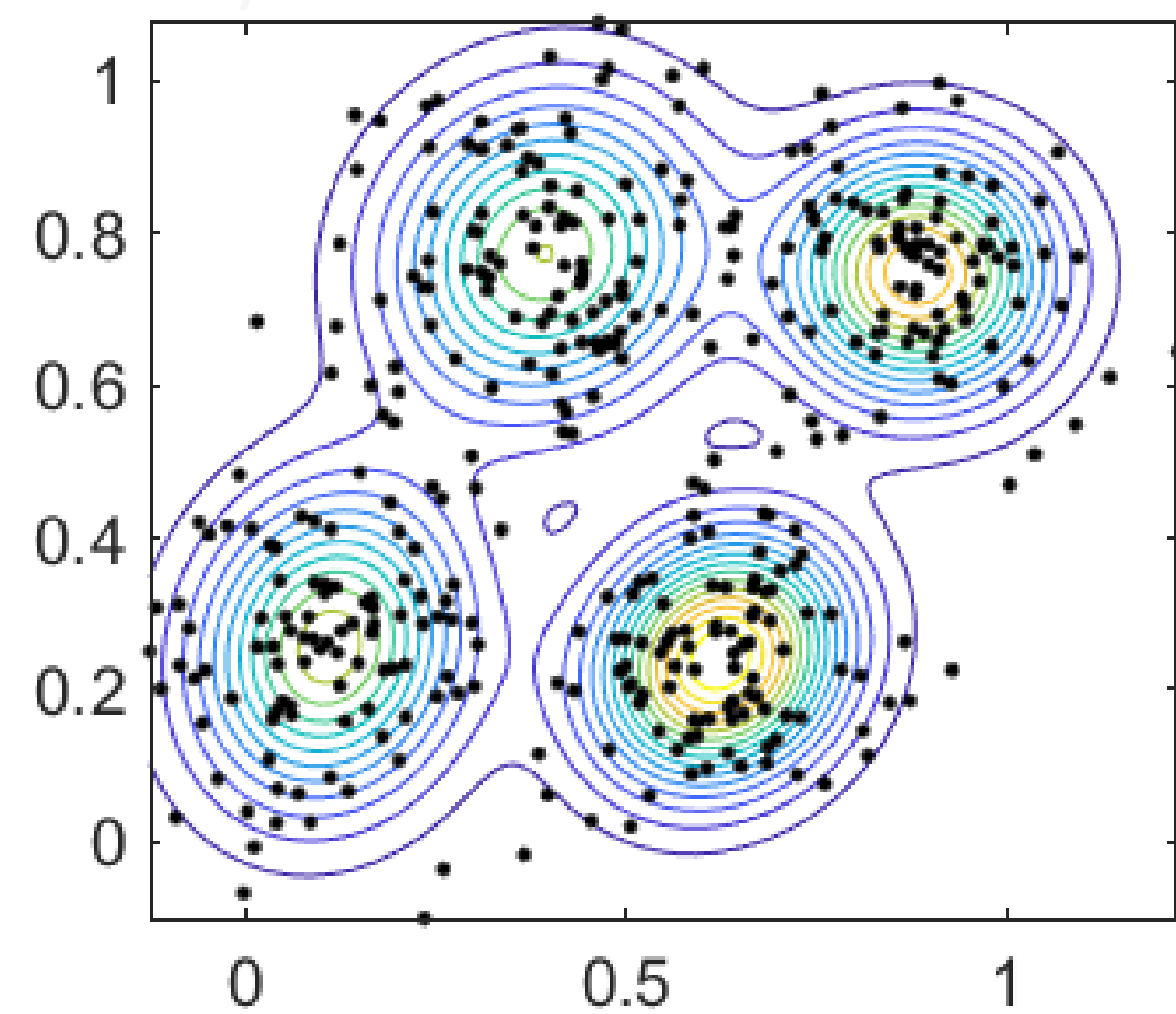
Let's generate a new dataset and run K-Means



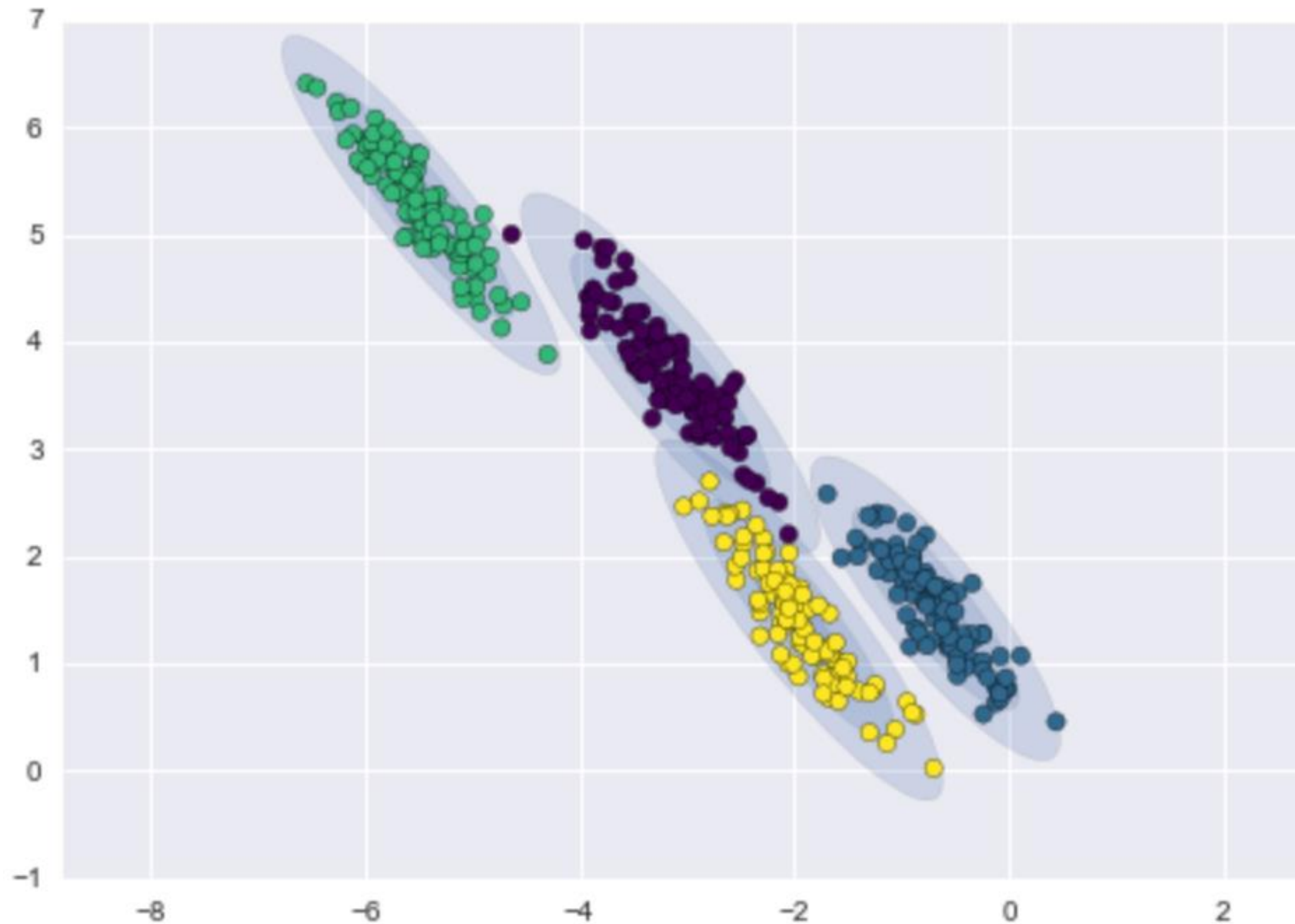
Let's run GMM on the first dataset



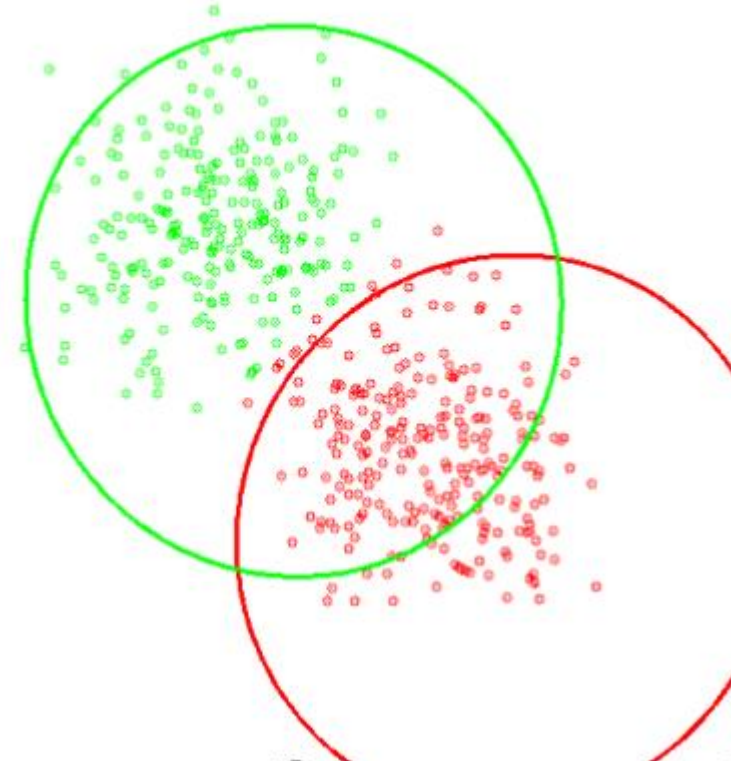
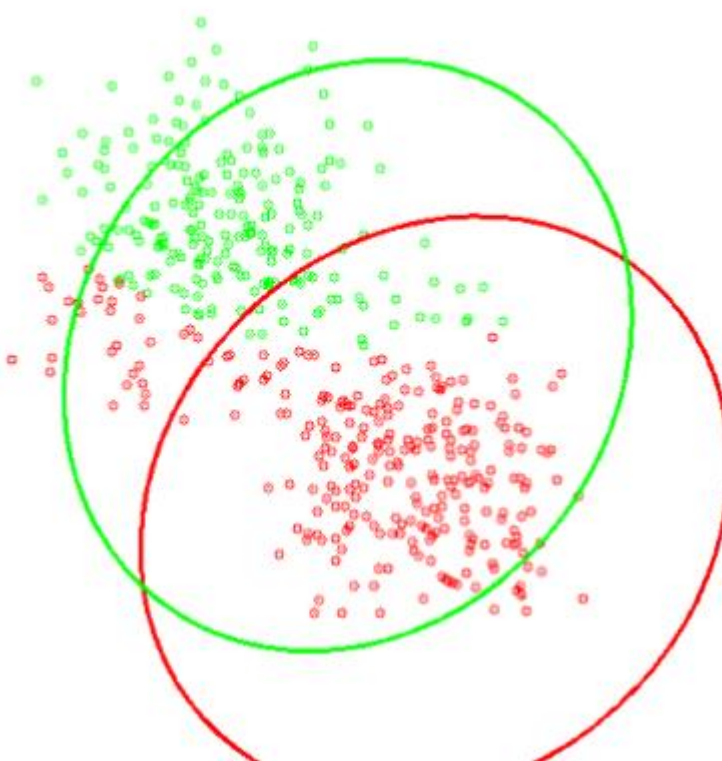
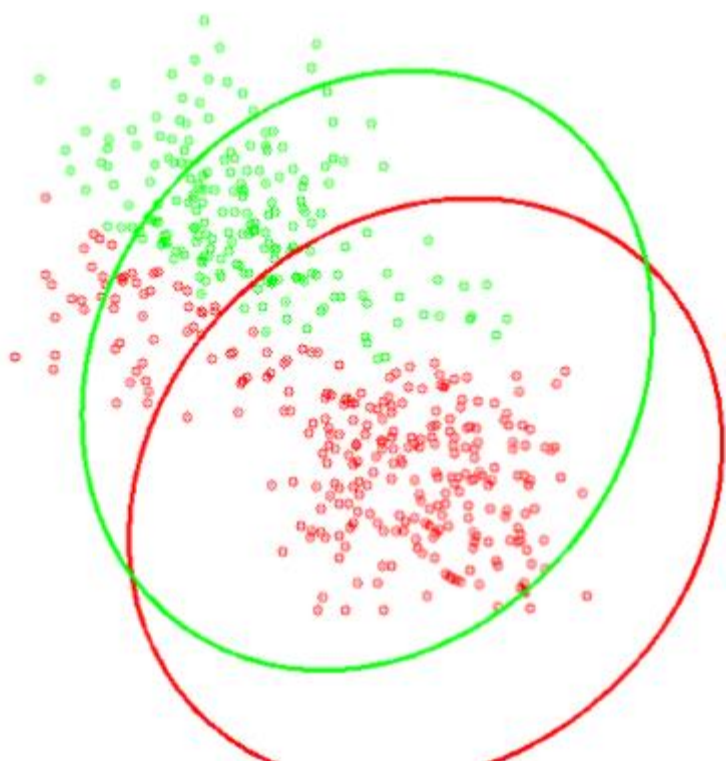
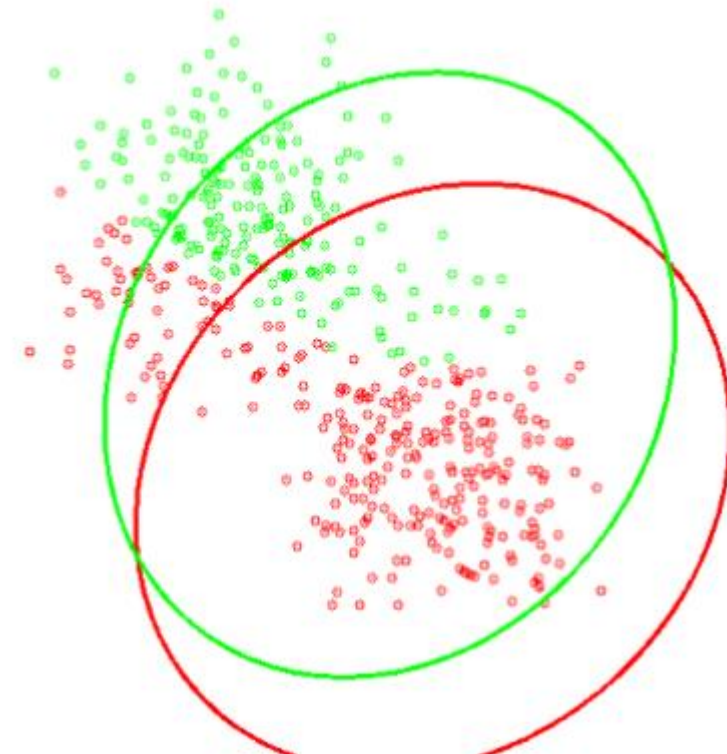
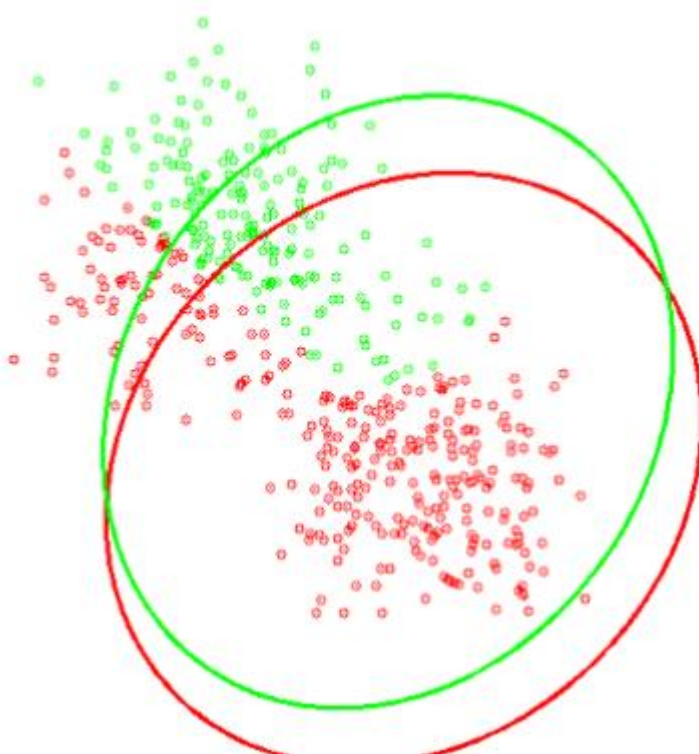
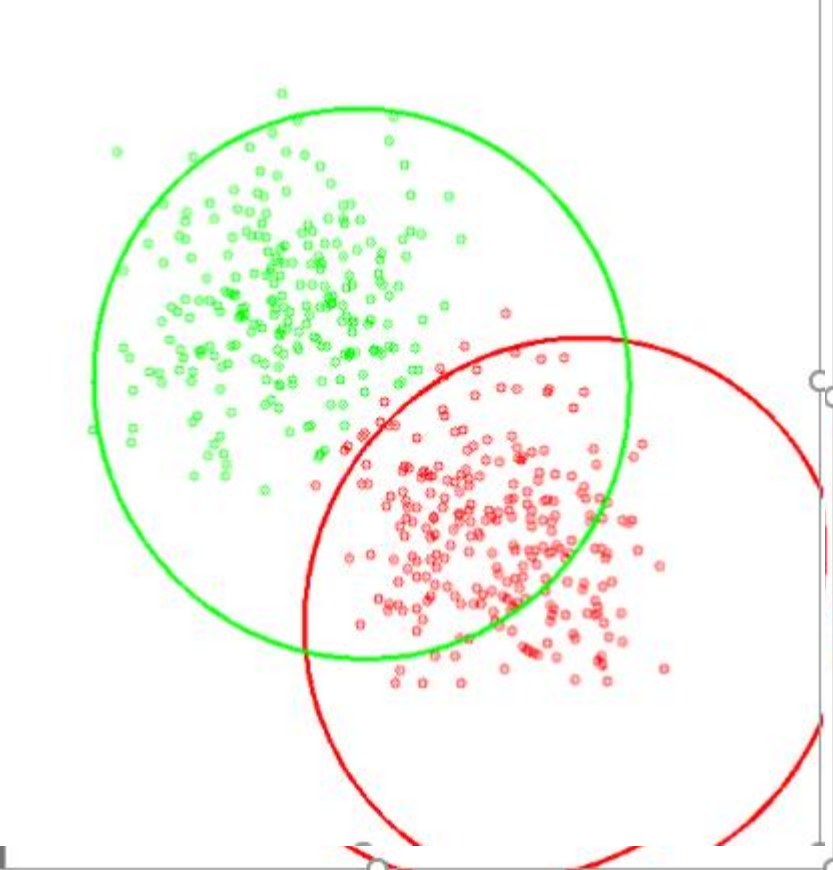
- How many parameters to optimize for each cluster/component?
- For this classification, what would be the offdiagonal part of covariance matrix? What about the diagonal?



Let's do GMM on the second dataset

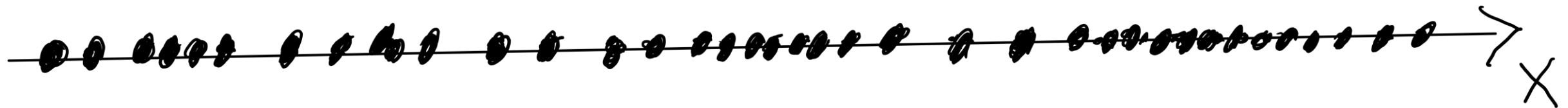
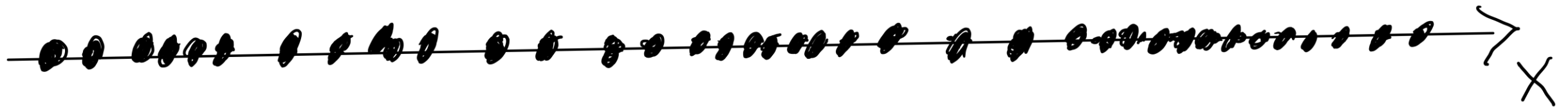
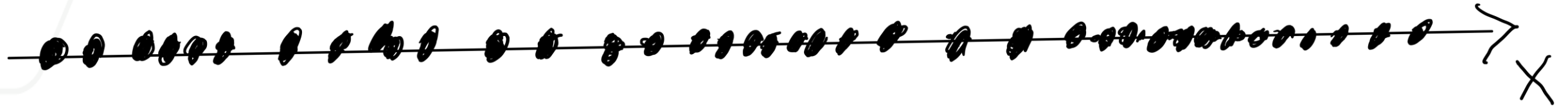


- What helped us adjust the shape of cluster/component?
- More parameters = More degree of flexibility
- Number of points in a cluster matters for probability.
- π_k = number of points in cluster/component k by total number of points

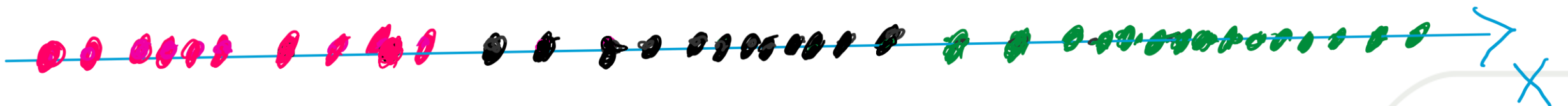
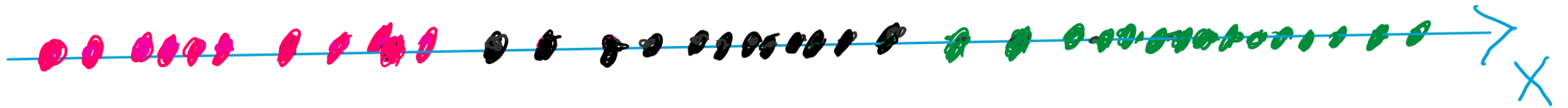




- cluster 1
- cluster 2
- cluster 3



Final Step



Mixture perspective – soft assignment

Let's create a **SINGLE** pdf that combines all three Gaussians!!!!

$$p(x) = \pi_0 N(X|\mu_0, \sigma_0) + \pi_1 N(X|\mu_1, \sigma_1) + \pi_2 N(X|\mu_2, \sigma_2)$$

Mixture perspective – Initialization

$$p(x) = \pi_0 N(X|\mu_0, \sigma_0) + \pi_1 N(X|\mu_1, \sigma_1) + \pi_2 N(X|\mu_2, \sigma_2)$$

Initial Step



Final Step



Maximizing Likelihood $\{\pi, \mu, \sigma\} \in \theta$

$$\begin{aligned} p(x|\pi, \mu, \sigma) &= p(x|\theta) = p(x) \\ &= \pi_0 N(X|\mu_0, \sigma_0) + \pi_1 N(X|\mu_1, \sigma_1) + \pi_2 N(X|\mu_2, \sigma_2) \end{aligned}$$

Math Notations: Hidden Variables and Responsibility

$$p(x) = \pi_0 N(X|\mu_0, \sigma_0) + \pi_1 N(X|\mu_1, \sigma_1) + \pi_2 N(X|\mu_2, \sigma_2)$$

$$P(z_k) = \pi_k$$

what is contribution of component k.

$$P(x|z_k) = N(x|\mu_k, \sigma_k)$$

given that we are in component k, what is the probability of x.

$$P(z_k|x^{(i)}) = \textit{responsibility}$$

given we are looking at datapoint $x^{(i)}$ what is the probability of that datapoint belonging to component k

Math Notations: Hidden Variables and Responsibility

$$P(z_k) = \pi_k$$

$$P(x|z_k) = N(x|\mu_k, \sigma_k)$$

$$P(z_k|x^{\{i\}}) =$$

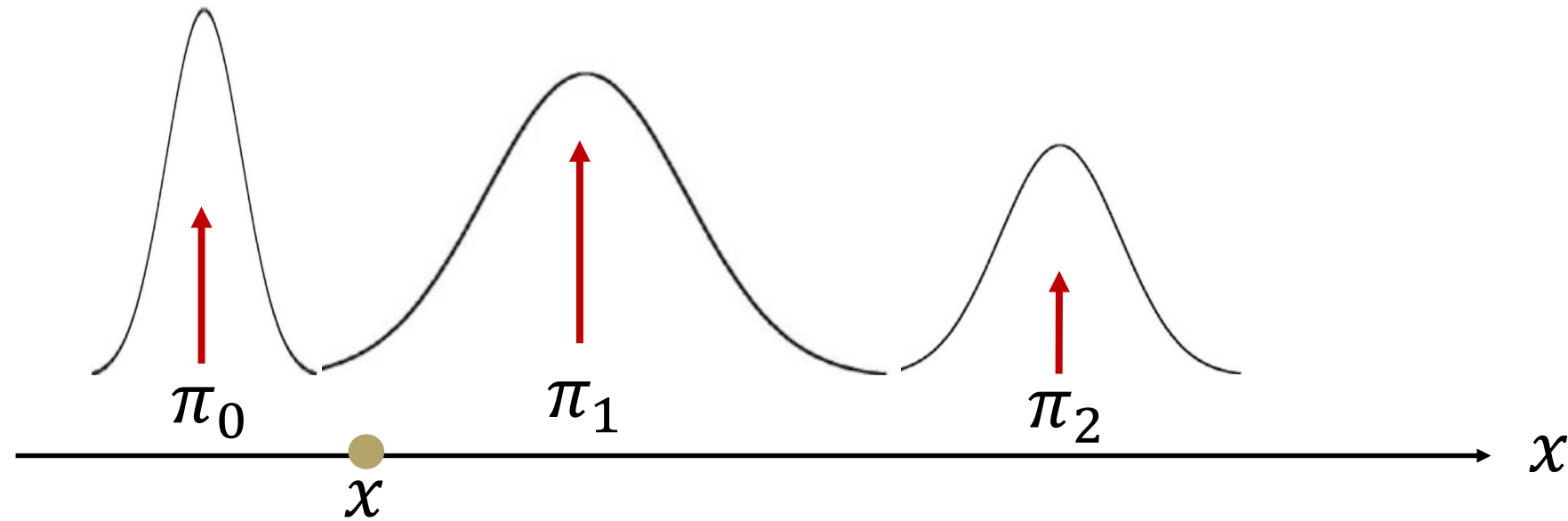
Recap

- GMM/Soft Clustering
- Responsibility Matrix
- $P(z_k) =$
- $P(x|z_k) =$
- Computing $P(z_k|x)$

$$p(x) = \pi_0 N(X|\mu_0, \sigma_0) + \pi_1 N(X|\mu_1, \sigma_1) + \pi_2 N(X|\mu_2, \sigma_2)$$

How can we find the probability of each data point in each cluster?

What is soft assignment?



What is the probability of a datapoint x in each component?

How many components we have here? **3**

How many probability? **3**

What is the sum value of the 3 probabilities for each datapoint? **1**

Mixture Models

- Formally a Mixture Model is the weighted sum of a number of pdfs where the weights are determined by a distribution,

π

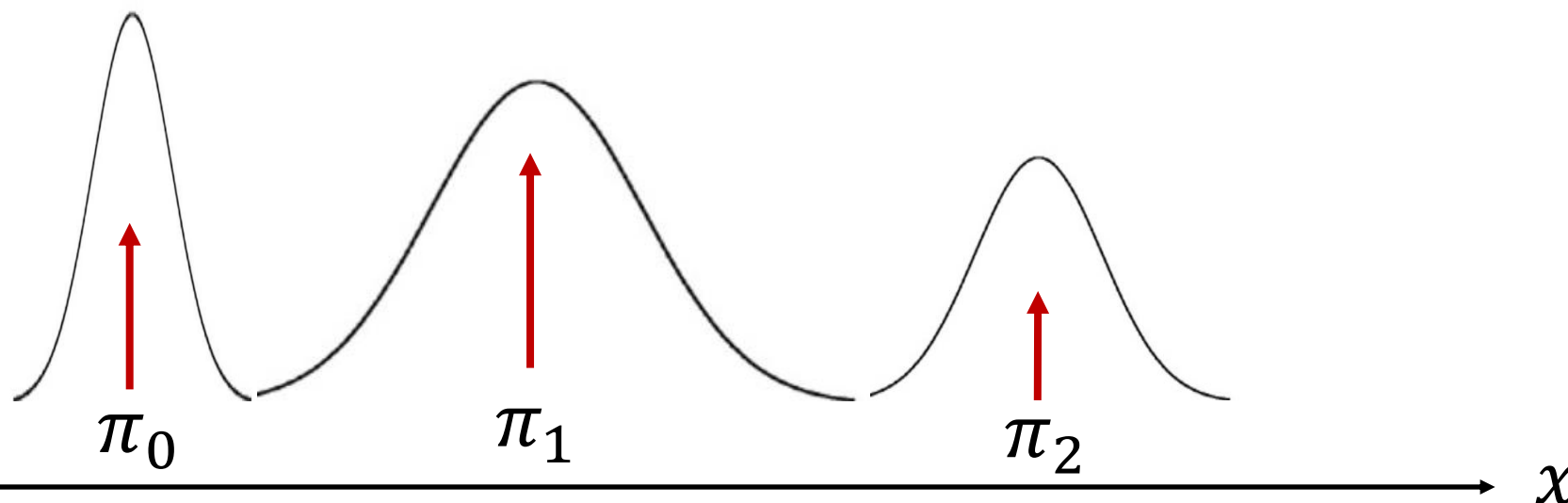
$$p(x) = \pi_0 f_0(x) + \pi_1 f_1(x) + \pi_2 f_2(x) + \dots + \pi_k f_k(x)$$

where $\sum_{i=0}^k \pi_i = 1$

$$p(x) = \sum_{i=0}^k \pi_i f_i(x)$$



What is **f** in GMM?



Mixture Models are Generative

- Generative simply means dealing with joint probability $p(x, z)$

$$p(x) = \pi_0 f_0(x) + \pi_1 f_1(x) + \cdots + \pi_k f_k(x)$$

Let's say $f(\cdot)$ is a **Gaussian** distribution

$$p(x) = \pi_0 N(X|\mu_0, \sigma_0) + \pi_1 N(X|\mu_1, \sigma_1) + \cdots + \pi_k N(X|\mu_k, \sigma_k)$$

$$p(x) = \sum_k N(x|\mu_k, \sigma_k) \pi_k$$

$$p(x) = \sum_k p(x|z_k) p(z_k) \quad z_k \text{ is component } k$$

$$p(x) = \sum_k p(x, z_k)$$

Inferring Cluster Membership

- We have representations of the joint $p(x, z_{nk}|\theta)$ and the marginal, $p(x|\theta)$
- The conditional of $p(z_{nk}|x, \theta)$ can be derived using Bayes rule.
 - The **responsibility** that **a** mixture component takes for explaining an observation x .

$$\begin{aligned}\tau(z_k) = p(z_k|x) &= \frac{p(z_k)p(x|z_k)}{\sum_{j=1}^K p(z_j)p(x|z_j)} \\ &= \frac{\pi_k N(x|\mu_k, \Sigma_k)}{\sum_{j=1}^K \pi_j N(x|\mu_j, \Sigma_j)}\end{aligned}$$

Why having “Latent variable”

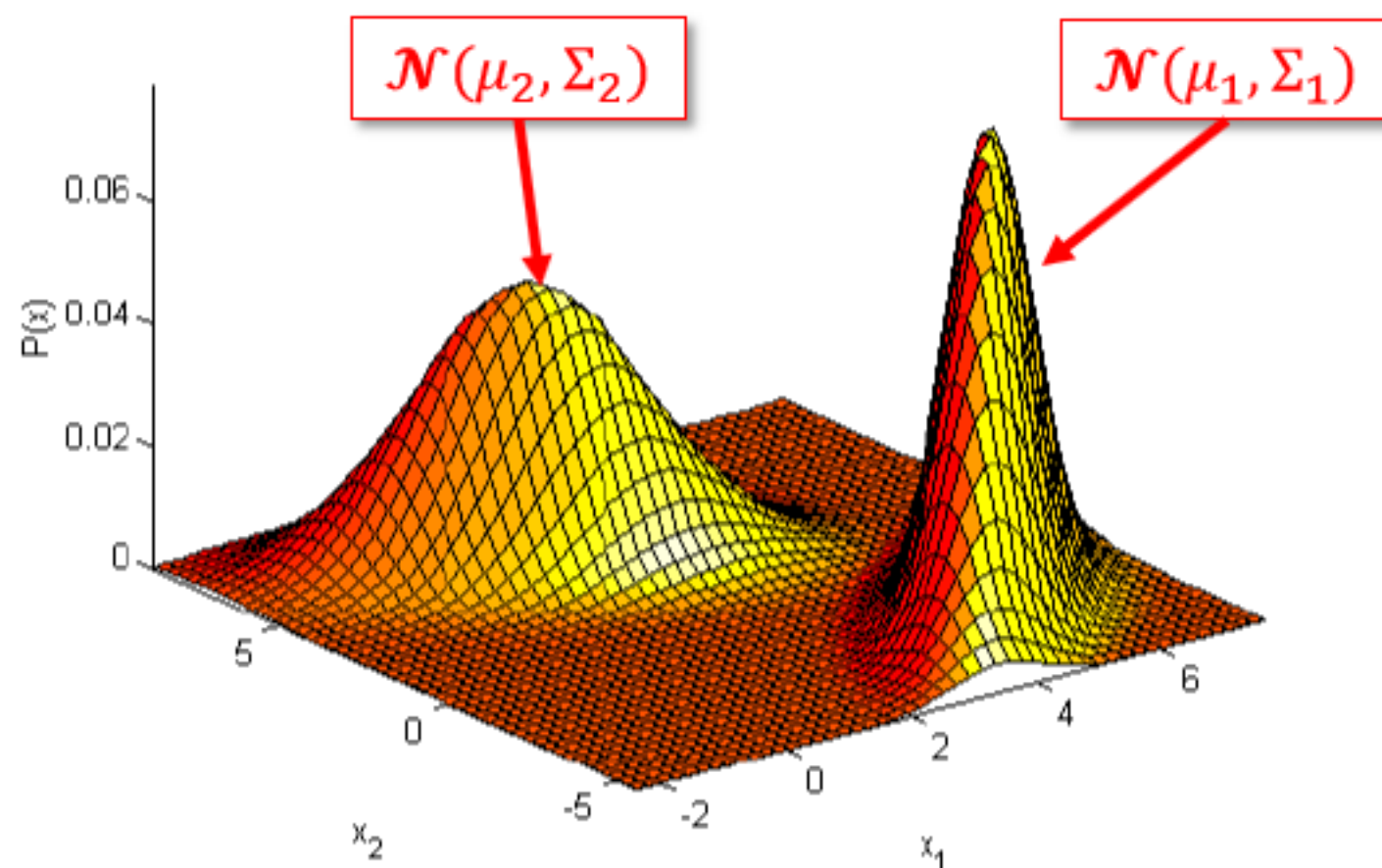
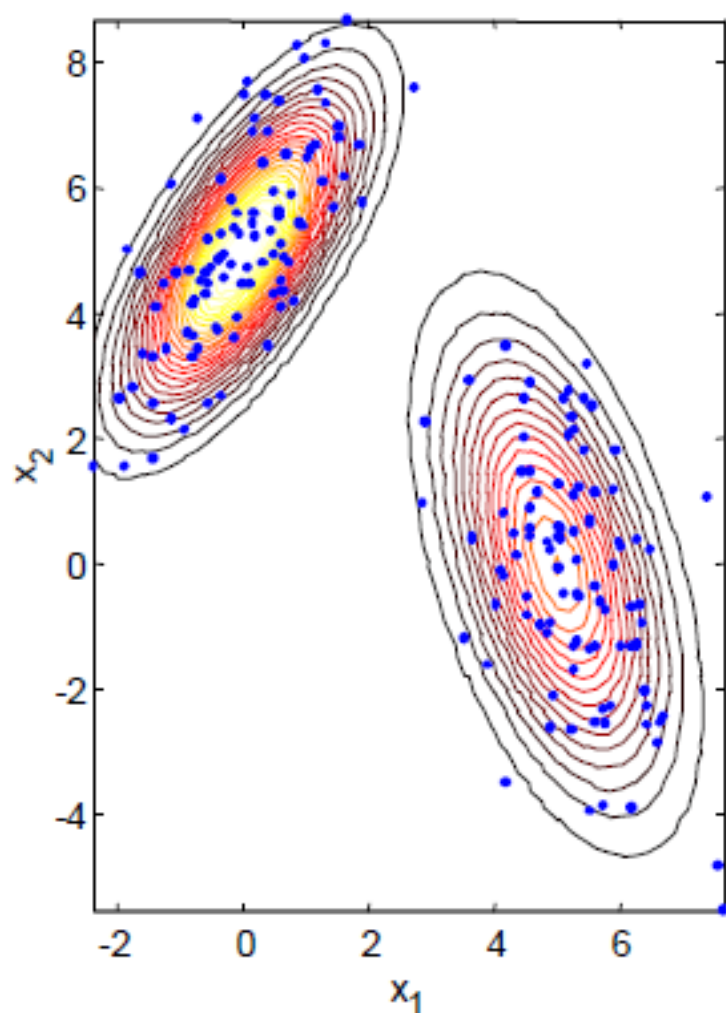
- A variable can be unobserved (latent) because:
 - it is an imaginary quantity meant to provide some simplified and abstractive view of the data generation process.
 - e.g., speech recognition models, mixture models (soft clustering)...
 - it is a real-world object and/or phenomena, but difficult or impossible to measure
 - e.g., the temperature of a star, causes of a disease, evolutionary ancestors ...
 - it is a real-world object and/or phenomena, but sometimes wasn't measured, because of faulty sensors, etc.
- Discrete latent variables can be used to partition/cluster data into sub-groups.
- Continuous latent variables (factors) can be used for dimensionality reduction (factor analysis, etc).

Quick Knowledge Check

- What does $P(x | z_k)$ represent in a GMM? A) The Normal distribution PDF for cluster k B) The responsibility of cluster k for data point x
- What does $P(z_k | x)$ represent in a GMM? A) The *responsibility* of cluster k for data point x . B) The *Normal distribution PDF* for cluster k
- What does $P(z_k)$ represent in a GMM? A) Cluster membership prior (mixture weight π_k) B) Responsibility of cluster k
- Which probability value is the final output we want after running the GMM? A) $P(z_k | x)$. B) $P(x | z_k)$
- For each cluster in a GMM, how many parameters do we need to randomly initialize? A) Two — μ_k and σ_k B) Three — μ_k , σ_k , and π_k

How about GMM for multimodal distribution?

- What if we know the data consists of a few Gaussians
- What if we want to fit parametric models



Gaussian Mixture Model

- A density model $p(X)$ may be multi-modal: model it as a mixture of uni-modal distributions (e.g. Gaussians)

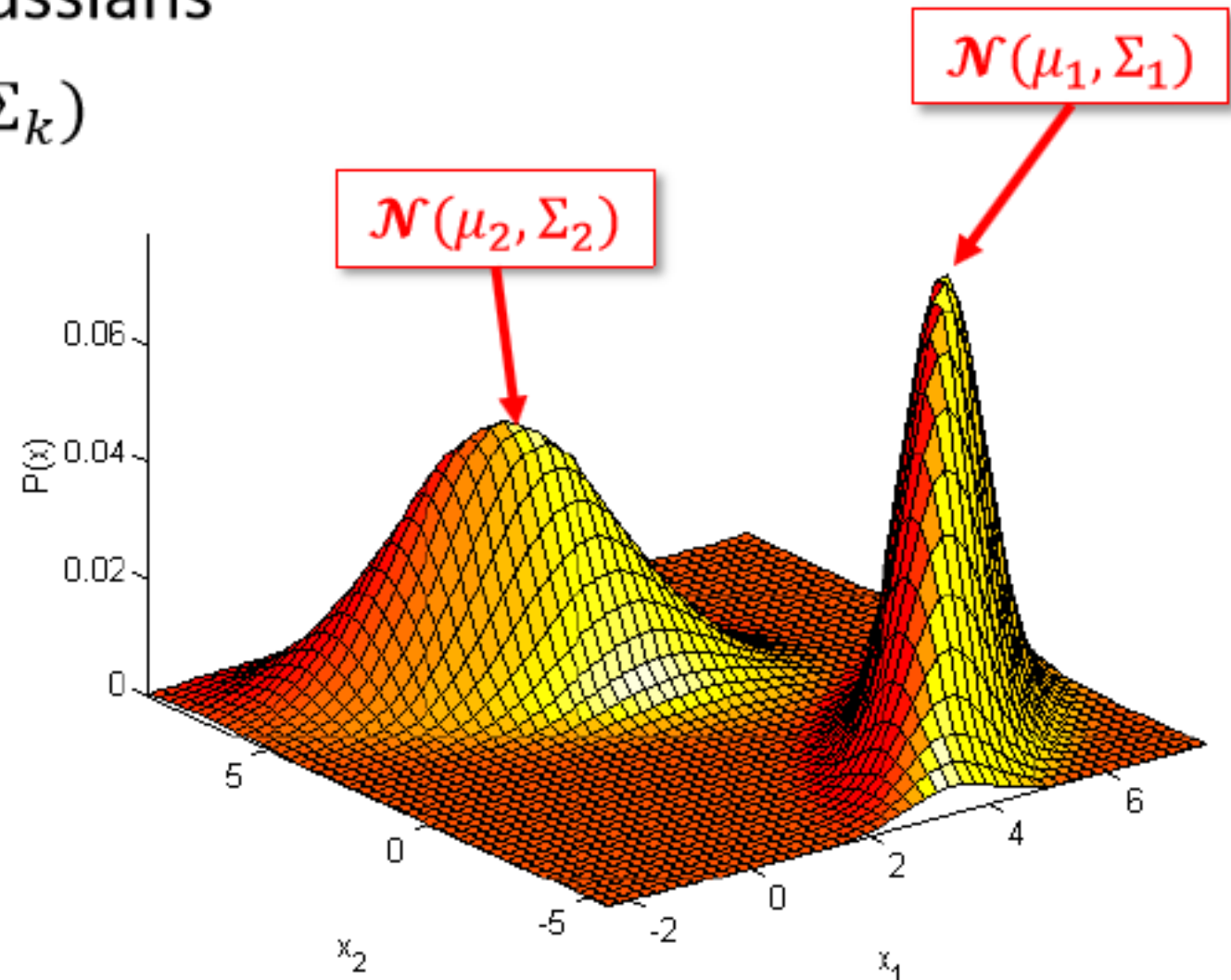
- Consider a mixture of K Gaussians

- $p(X) = \sum_{k=1}^K \pi_k \mathcal{N}(X|\mu_k, \Sigma_k)$

mixing
proportion

mixture
Component

- Learn $\pi_k \in (0,1), \mu_k, \Sigma_k$;



What are GMM parameters?

Mean μ_k

Variance σ_k

Size π_k

Marginal probability distribution

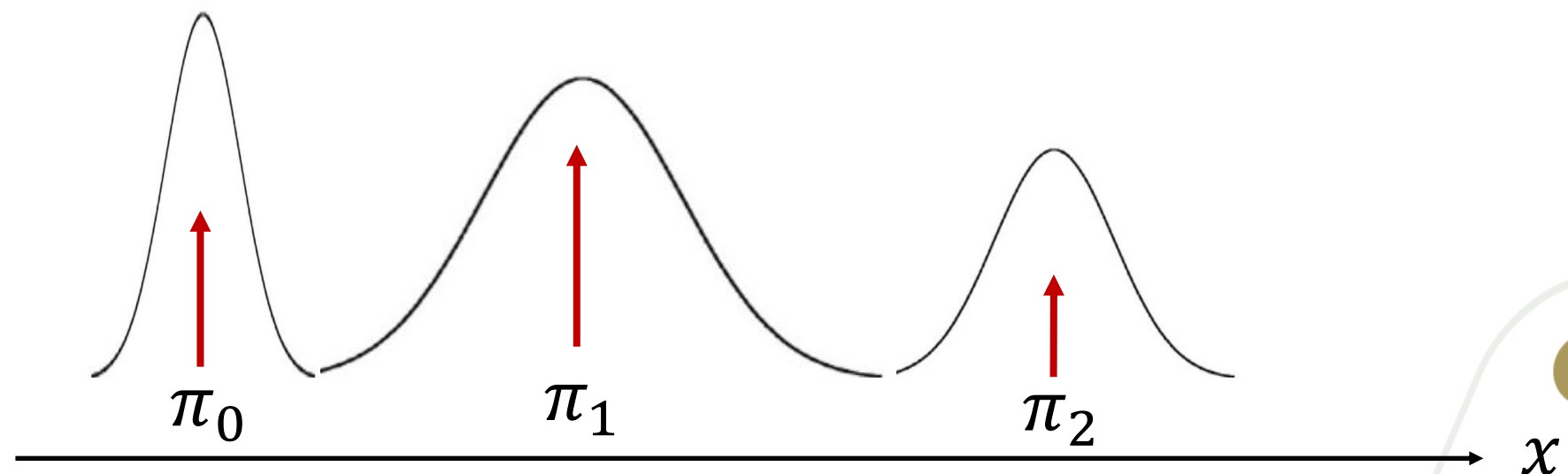
$$p(x|\theta) = \sum_k p(x, z_k|\theta) = \sum_k \underbrace{p(x|z_k, \theta)}_{f_k(x)} \underbrace{p(z_k|\theta)}_{\pi_k} = \sum_k N(x|\mu_k, \sigma_k) \pi_k$$

$$p(z_k|\theta) = \pi_k$$

Select a mixture component with probability π

$$p(x|z_k, \theta) = N(x|\mu_k, \sigma_k)$$

Sample from that component's Gaussian



Parameters' definition

- Purpose: GMM is a clustering algorithm derived from probabilistic theory that uses soft-assignment, meaning that data points have probability of being associated/generated from K gaussians/clusters. This is as opposed to K-means where data points definitively are either from a cluster or they're not.
- Gaussian Parameters
 - μ : Mean of each gaussian, can be compared to the K-means cluster centers
 - Σ : Covariance matrix of each gaussian, which represents how dimensions vary between each other. If it's the covariance of a specific dimension with itself, it is just the standard deviation of that variable/dimension. This is in a $D \times D$ matrix (for each gaussian) and every element $\Sigma_{i,j}$ represents the covariance of dimension i with j . If you assume the dimensions are independent, then only the diagonals are non-zero.
 - z_{nk} : Latent variable which isn't explicitly known, but tells us which gaussian each datapoint was generated from. z is binary, it either's 1 (point x came from gaussian k) or 0 (point x did not come from gaussian k)
 - $p(z_{nk}) = \pi$: Mixing proportions/weights, which represent the fraction of data points that are generated from/associated with each gaussian. These sum to 1.
- $N(X_n | \mu_k, \Sigma_k)$: This term is the probability of some data point X_n occurring based on the assumption that it is generated from gaussian k . Mathematically, this is equal to the likelihood ($p(x|z, \mu_k, \Sigma_k)$). Multiply this with the mixing weight π , and you get the joint distribution $p(x=X_n, z=k)$.
- $P(X)$: If you sum up all the $N(X_n | \mu_k, \Sigma_k) * \pi$ terms for each cluster, you get the probability of the entire data set occurring.
- $\gamma(z_{nk})$ or $P(z_{nk} | x_n)$: We call this term the "responsibility." It is the probability of z for "A" data point x , meaning that this is probability that point n is generated from gaussian k normalized by $P(X)$, the probability of the entire data set occurring.

Well, we don't know π_k, μ_k, Σ_k
What should we do?

We want
$$\tau(z_k) = \frac{\pi_k N(x|\mu_k, \Sigma_k)}{\sum_{j=1}^K \pi_j N(x|\mu_j, \Sigma_j)}$$

We use “Maximum Likelihood Estimation” (MLE) to solve the problem.

$$p(x) = p(x|\theta) = \sum_k p(x, z_k|\theta) = \sum_k p(z_k|\theta)p(x|z_k, \theta) = \sum_{k=0}^K \pi_k N(x|\mu_k, \Sigma_k)$$

Let's identify a likelihood function, why?

Because we use likelihood function to optimize the probabilistic model parameters!

$$\arg \max p(x|\theta) = p(x|\pi, \mu, \Sigma) = \prod_{i=1}^N p(x^{\{i\}}|\theta) = \prod_{i=1}^N \sum_{k=0}^K \pi_k N(x^{\{i\}}|\mu_k, \Sigma_k)$$

$$\arg \max p(x) = p(x|\pi, \mu, \Sigma) = \prod_{i=1}^N p(x^{\{i\}}|\theta) = \prod_{i=1}^N \sum_{k=0}^K \pi_k N(x^{\{i\}}|\mu_k, \Sigma_k)$$

$$\ln[p(x)] = \ln[p(x|\pi, \mu, \Sigma)]$$

- As usual: Identify a likelihood function

$$\ln p(x|\pi, \mu, \Sigma) = \sum_{i=1}^N \ln \sum_{k=0}^K \pi_k N(x^{\{i\}}|\mu_k, \Sigma_k)$$

- And set partials to zero...

This is **non-convex** and **very difficult to optimize directly** because:

- The clusters (components) are **unknown** — we don't know which cluster generated each point.
- There is no **closed-form solution** for the parameters since you can't just take derivatives and set them to zero easily.
- The presence of **latent variables** (the cluster assignments z) makes the problem incomplete.

Maximum Likelihood of a GMM

- Optimization of means.

$$\ln p(x|\pi, \mu, \Sigma) = \sum_{n=1}^N \ln \left\{ \sum_{k=1}^K \pi_k N(x_n | \mu_k, \Sigma_k) \right\}$$

$$\frac{\partial \ln p(x|\pi, \mu, \Sigma)}{\partial \mu_k} = \sum_{n=1}^N \frac{\pi_k N(x_n | \mu_k, \Sigma_k)}{\sum_j \pi_j N(x_n | \mu_j, \Sigma_j)} \Sigma_k^{-1} (x_k - \mu_k) = 0$$

$$= \sum_{n=1}^N \tau(z_{nk}) \Sigma_k^{-1} (x_k - \mu_k) = 0$$

$$\mu_k = \frac{\sum_{n=1}^N \tau(z_{nk}) x_n}{\sum_{n=1}^N \tau(z_{nk})}$$

Maximum Likelihood of a GMM

- Optimization of covariance

$$\ln p(x|\pi, \mu, \Sigma) = \sum_{n=1}^N \ln \left\{ \sum_{k=1}^K \pi_k N(x_n | \mu_k, \Sigma_k) \right\}$$

$$\Sigma_k = \frac{1}{\sum_{n=1}^N \tau(z_{nk})} \sum_{n=1}^N \tau(z_{nk}) (x_n - \mu_k)(x_n - \mu_k)^T$$

Maximum Likelihood of a GMM

- Optimization of mixing term

$$\ln p(x|\pi, \mu, \Sigma) + \lambda \left(\sum_{k=1}^K \pi_k - 1 \right)$$

$$0 = \sum_{n=1}^N \frac{N(x_n|\mu_k, \Sigma_k)}{\sum_j \pi_j N(x_n|\mu_j, \Sigma_j)} + \lambda$$

$$\pi_k = \frac{\sum_{n=1}^N \tau(z_{nk})}{N}$$

MLE of a GMM

$$\mu_k = \frac{\sum_{n=1}^N \tau(z_{nk}) x_n}{N_k}$$

$$\Sigma_k = \frac{1}{N_k} \sum_{n=1}^N \tau(z_{nk}) (x_n - \mu_k)(x_n - \mu_k)^T$$

$$\pi_k = \frac{N_k}{N}$$

$$N_k = \sum_{n=1}^N \tau(z_{nk})$$

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EM for GMMs

- E-step: Evaluate the Responsibilities

$$\tau(z_k) = \frac{\pi_k N(x_n | \mu_k, \Sigma_k)}{\sum_{j=1}^K \pi_j N(x_n | \mu_j, \Sigma_j)}$$

EM for GMMs

- M-Step: Re-estimate Parameters by optimizing objective function (maximizing log likelihood)

$$\mu_k^{new} = \frac{\sum_{n=1}^N \tau(z_{nk}) x_n}{N_k}$$

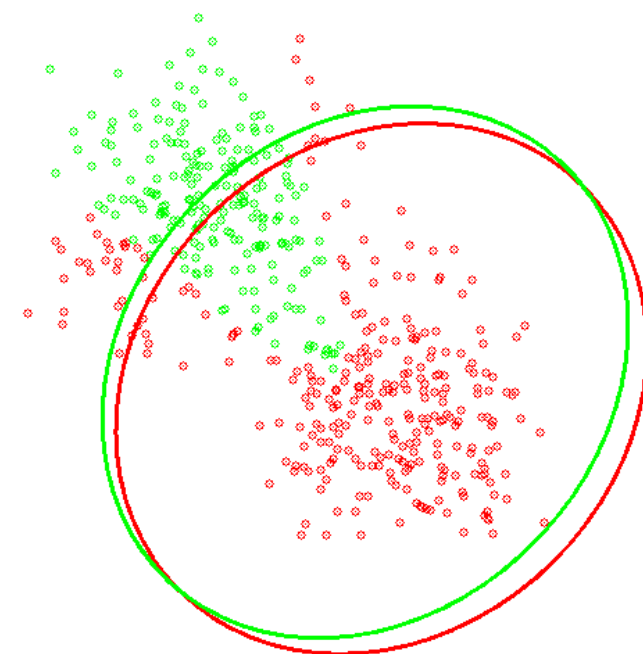
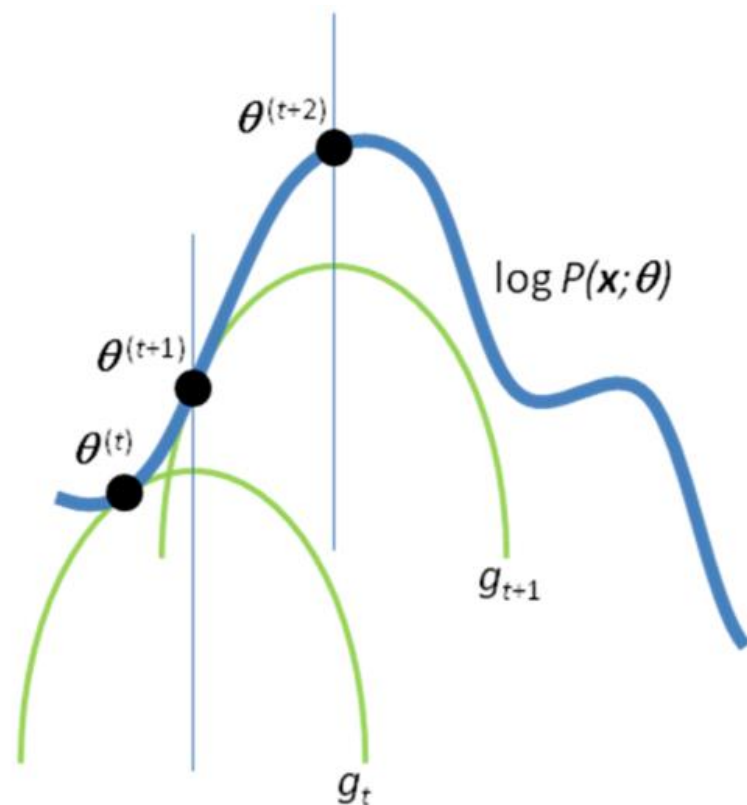
$$\Sigma_k^{new} = \frac{1}{N_k} \sum_{n=1}^N \tau(z_{nk}) (x_n - \mu_k^{new})(x_n - \mu_k^{new})^T$$

- Repeat E and M steps until convergence
- (until log likelihood does not change much and is hence maximized)

$$\pi_k^{new} = \frac{N_k}{N}$$

Expectation Maximization

- Expectation Maximization (EM) is a general algorithm to deal with hidden variables.
- Two steps:
 - E-Step: Fill-in hidden values using inference
 - M-Step: Apply standard MLE method to estimate parameters
- EM always converges to a local minimum of the likelihood.



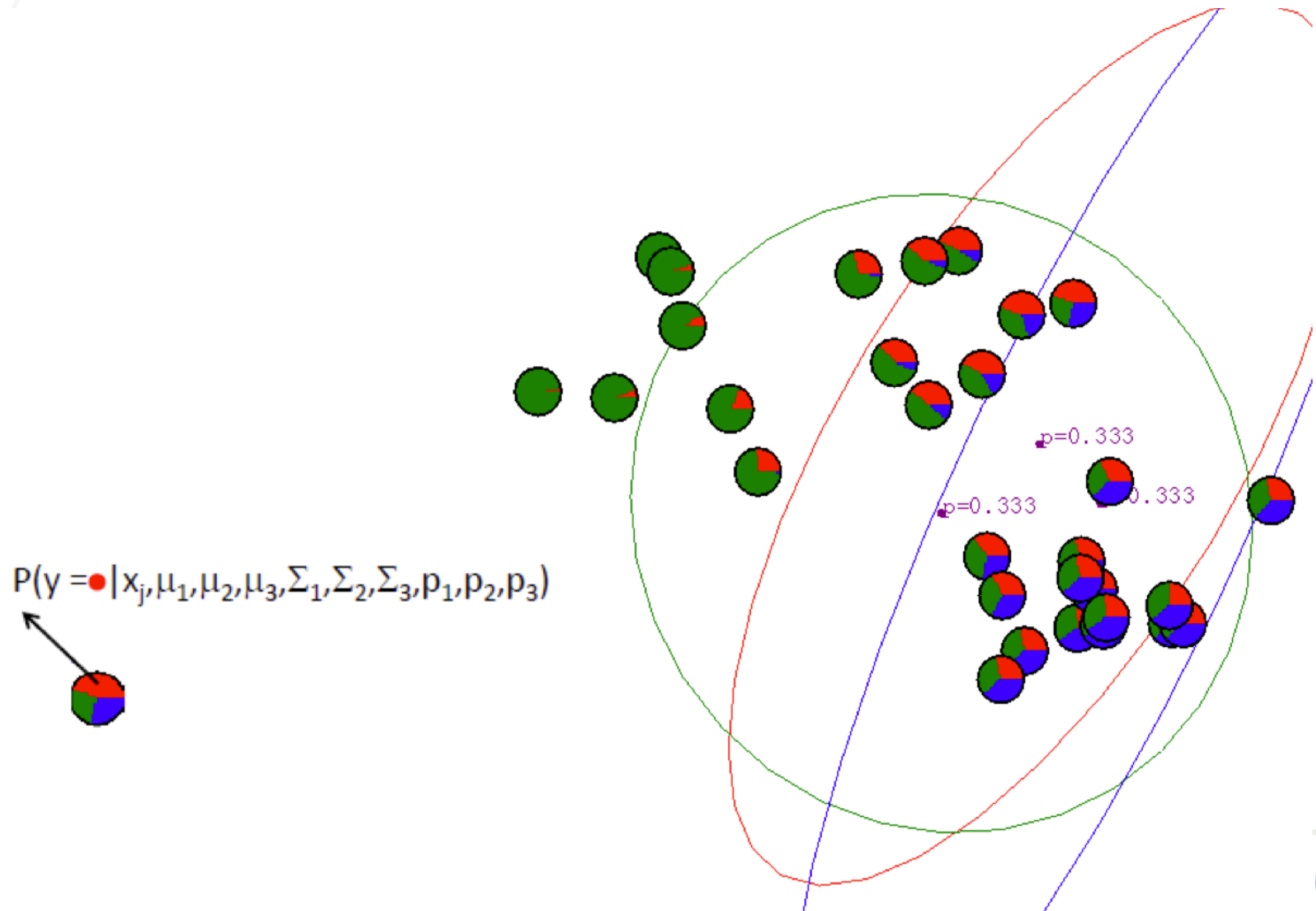
EM for Gaussian Mixture Model: Example

`covariance_type="diag"`

`covariance_type="spherical"`

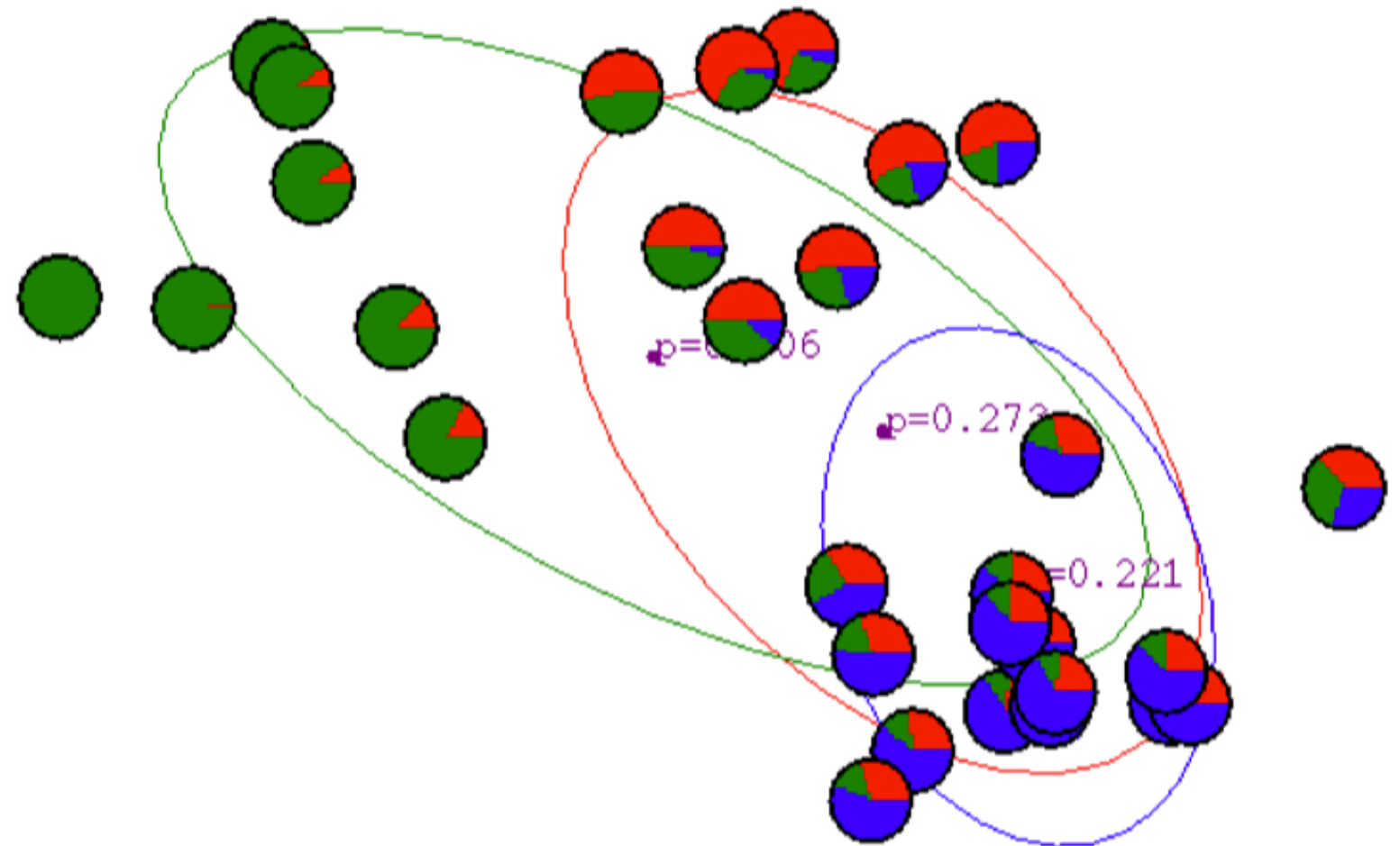
`covariance_type="full"`

EM for Gaussian Mixture Model:



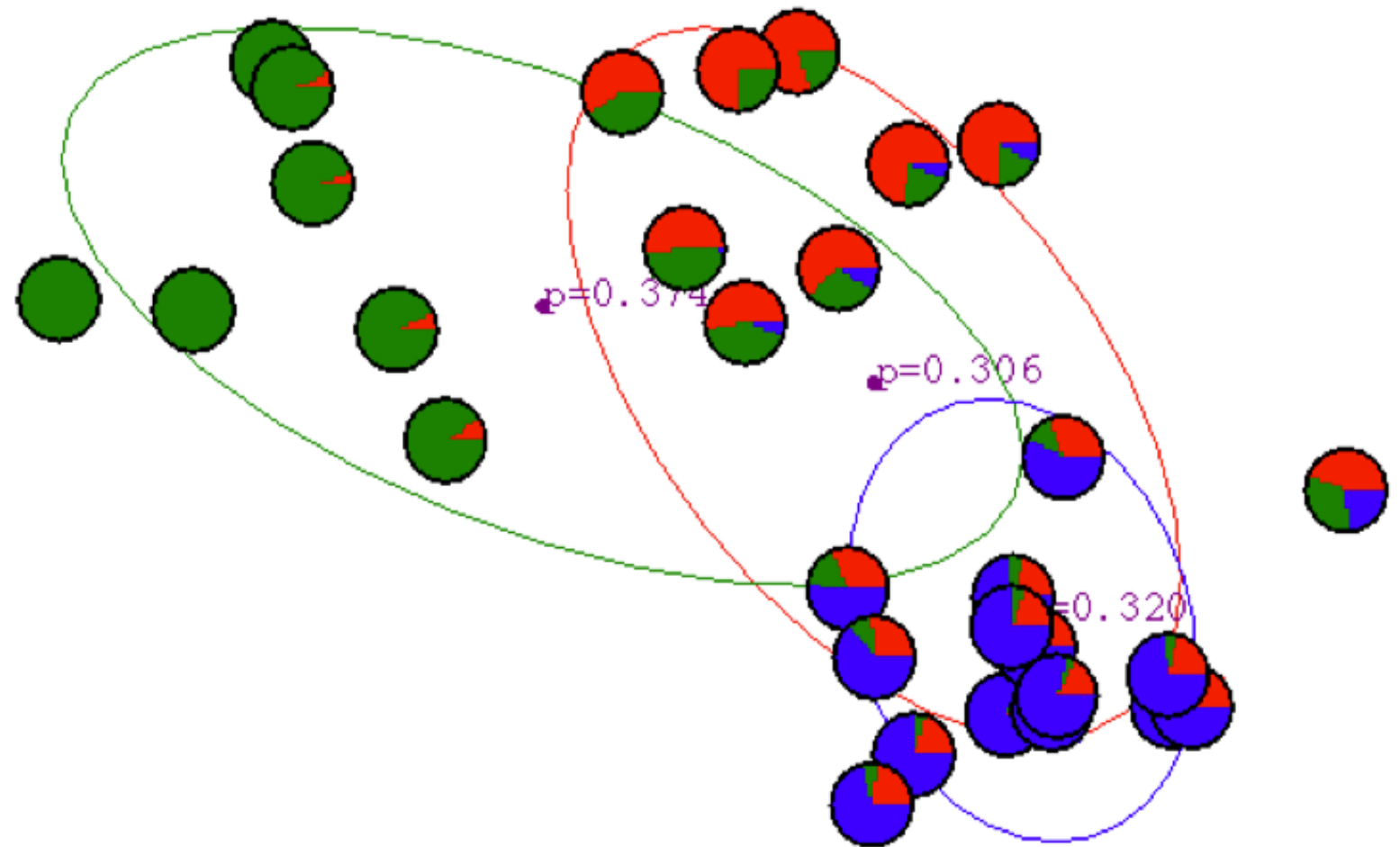
EM for Gaussian Mixture Model: Example

After 1st iteration



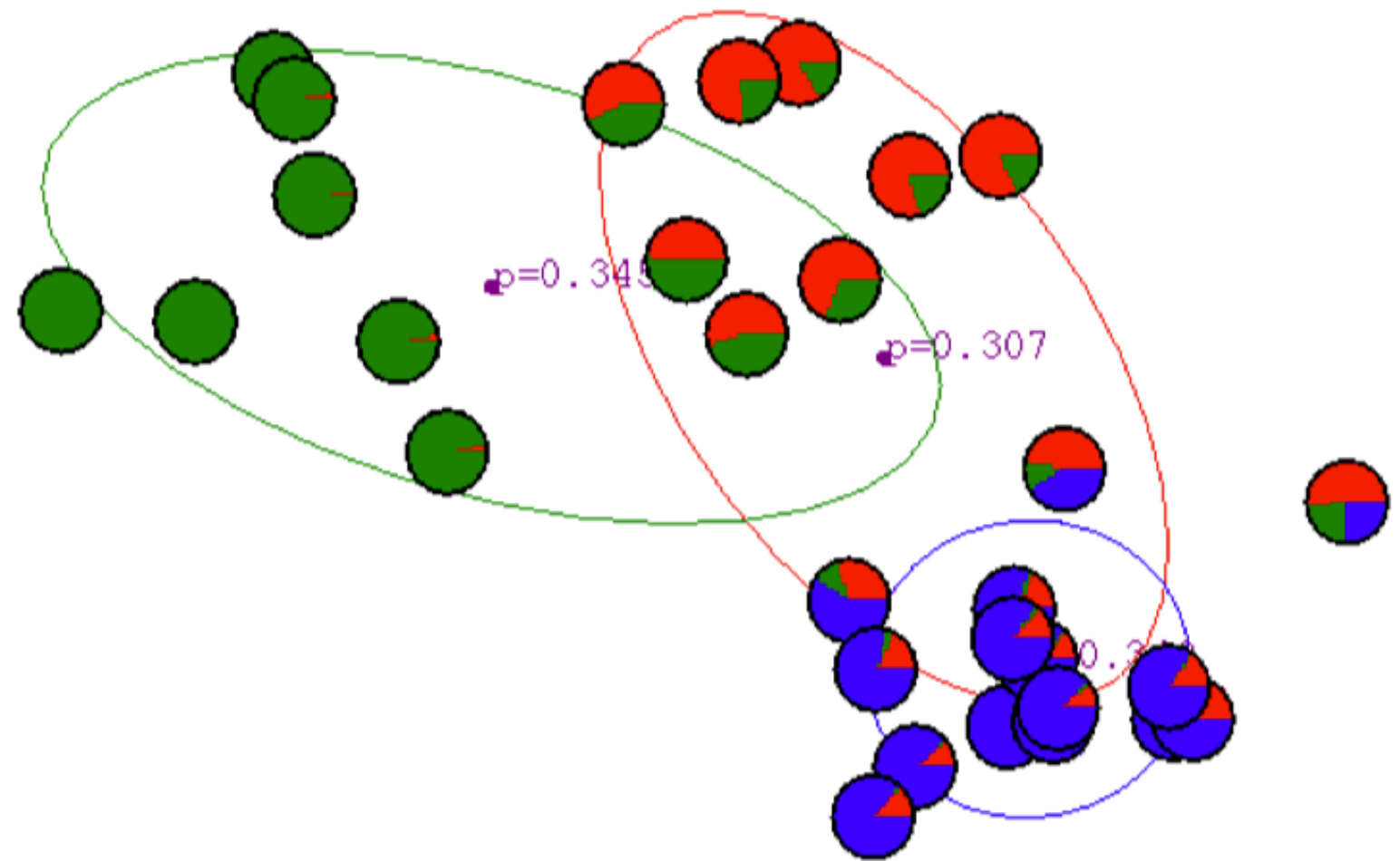
EM for Gaussian Mixture Model: Example

After 2nd iteration



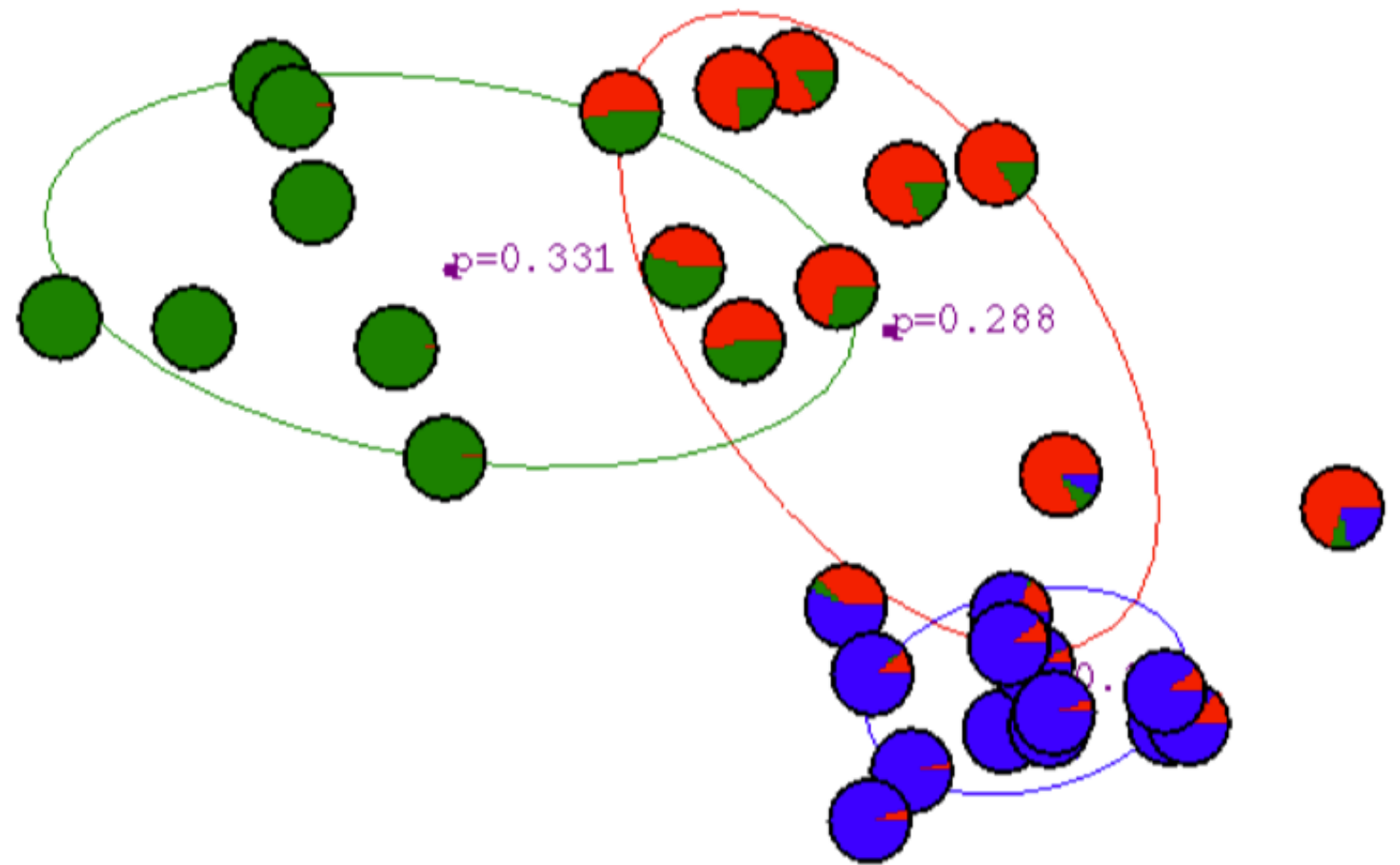
EM for Gaussian Mixture Model: Example

After 3rd iteration



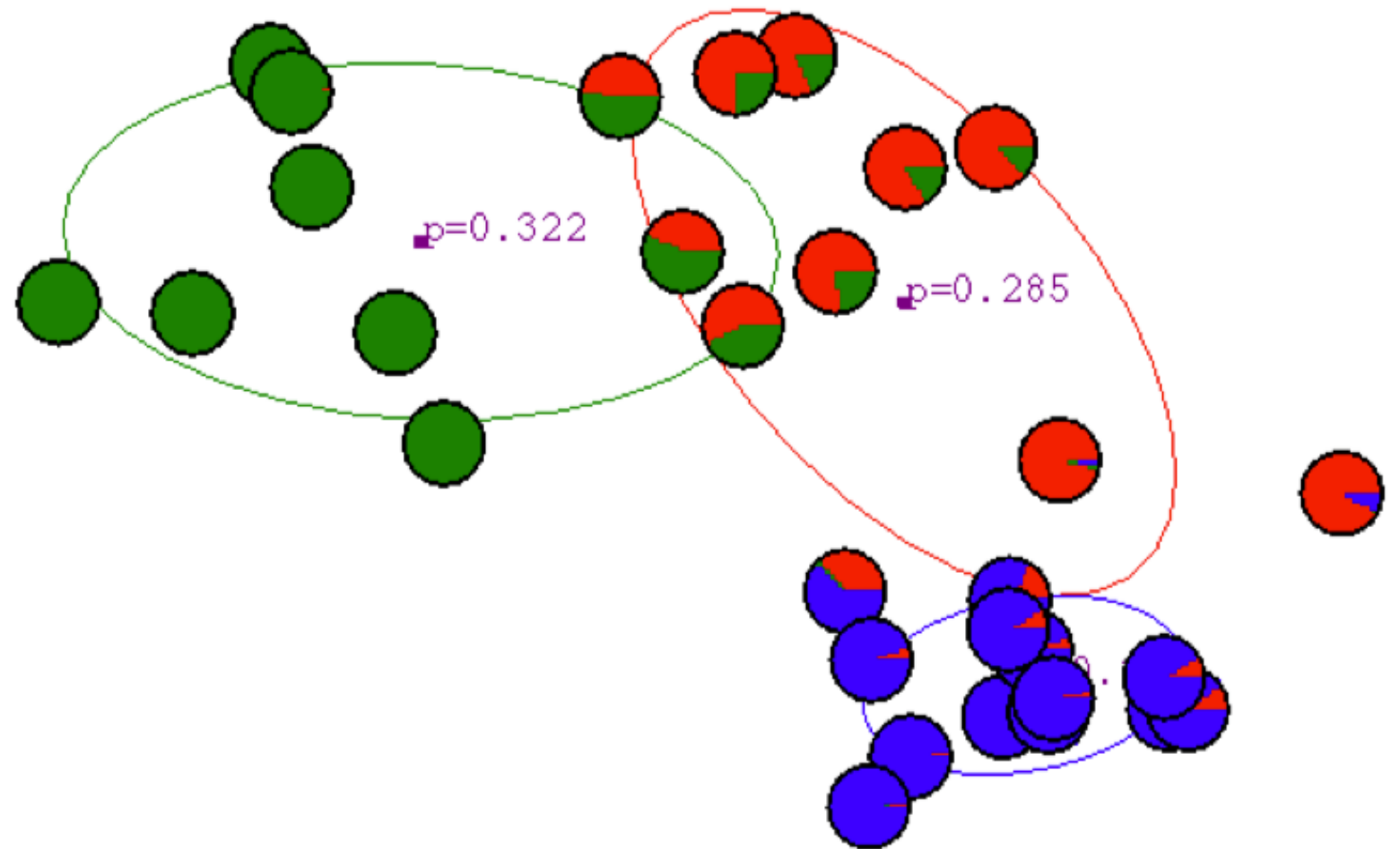
EM for Gaussian Mixture Model: Example

After 4th iteration



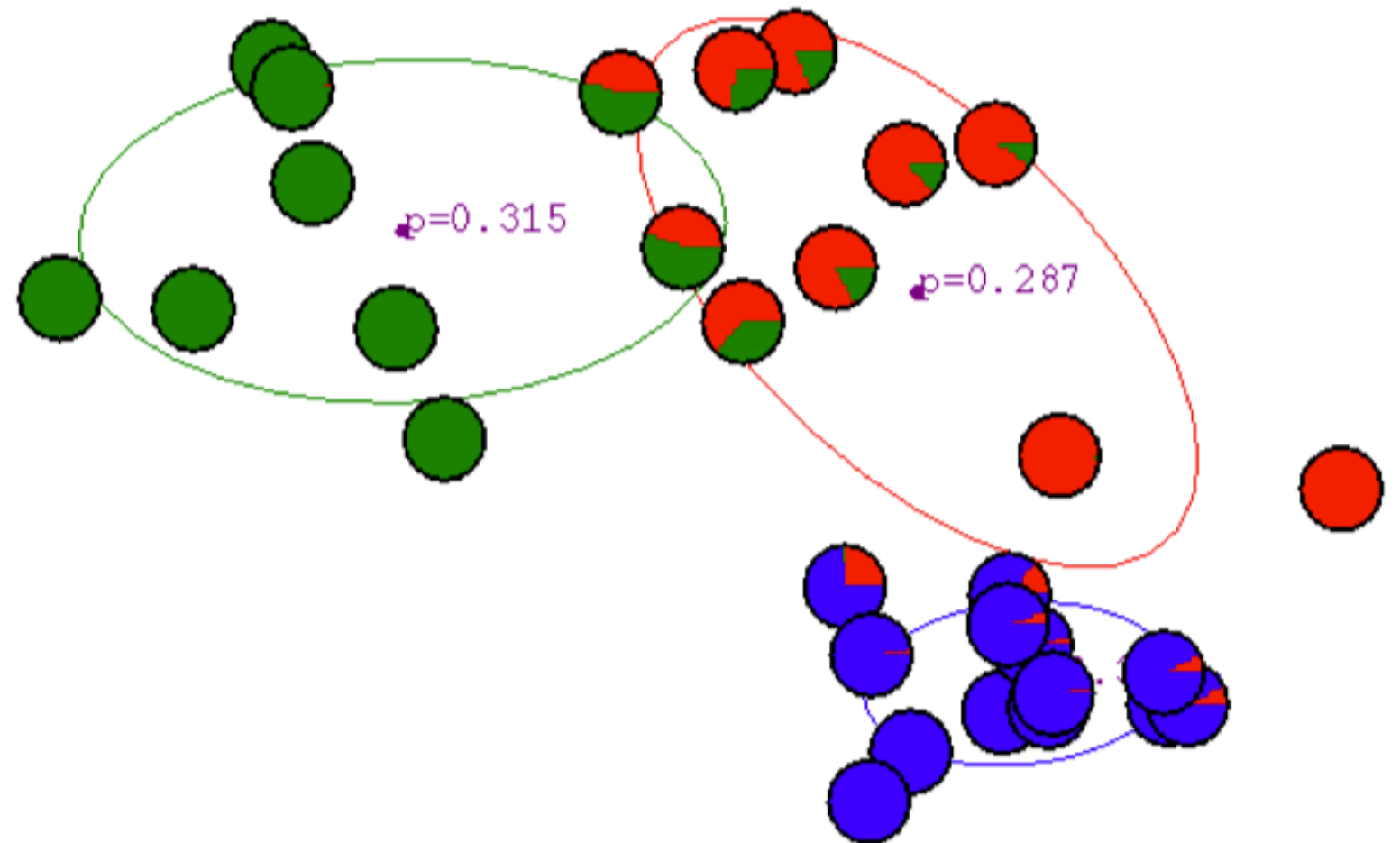
EM for Gaussian Mixture Model: Example

After 5th iteration



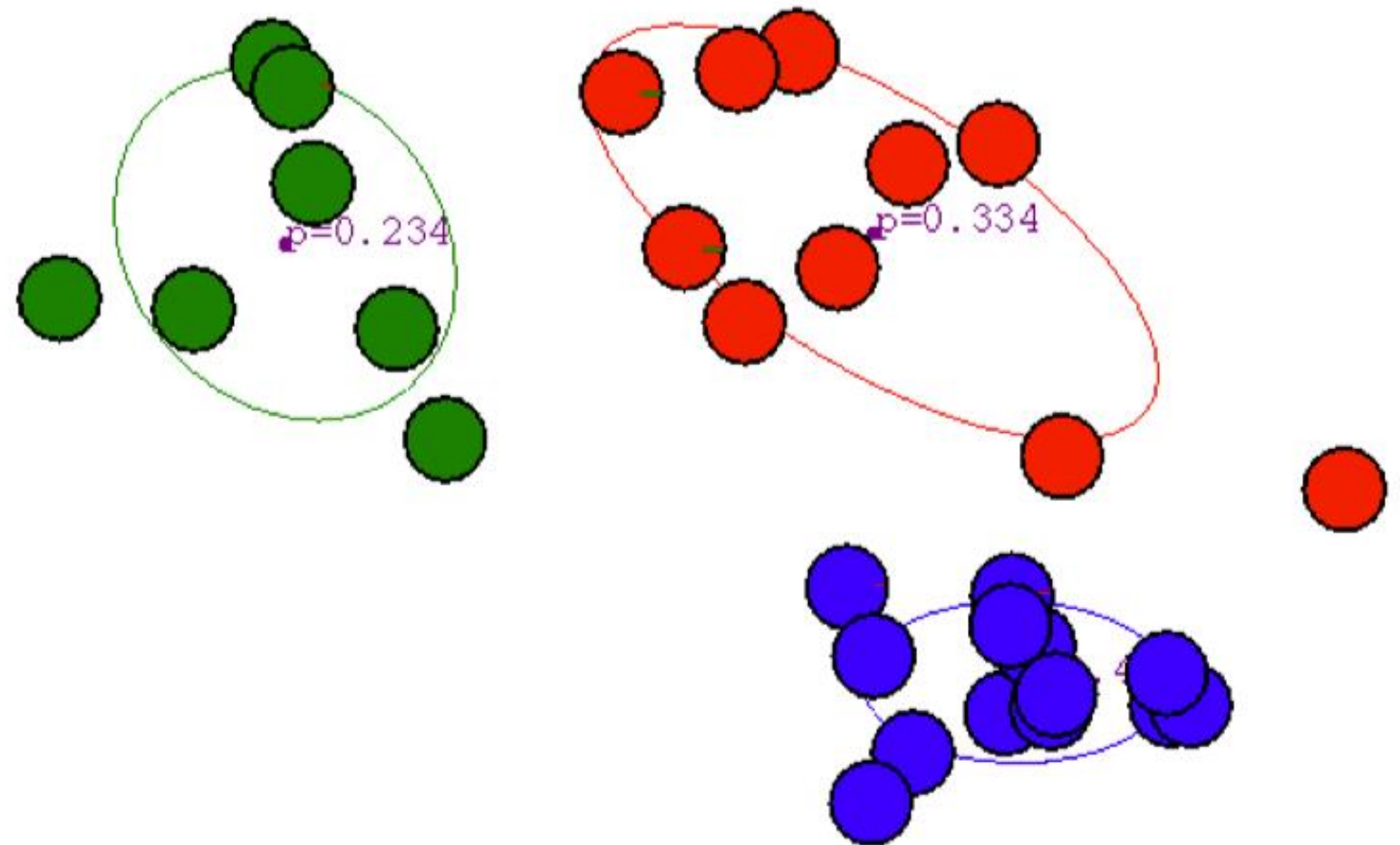
EM for Gaussian Mixture Model: Example

After 6th iteration



EM for Gaussian Mixture Model: Example

After 20th iteration



EM Algorithm for GMM (matrix form)

Given a Gaussian mixture model, the goal is to maximize the likelihood function with respect to the parameters comprising the means and covariances of the components and the mixing coefficients).

1. Initialize the means μ_j , covariances Σ_j and mixing coefficients π_j , and evaluate the initial value of the log likelihood.
2. **E step.** Evaluate the responsibilities using the current parameter values

$$\gamma(z_k) = \frac{\pi_k \mathcal{N}(\mathbf{x} | \mu_k, \Sigma_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(\mathbf{x} | \mu_j, \Sigma_j)}$$

EM Algorithm for GMM (matrix form)

3. **M step.** Re-estimate the parameters using the current responsibilities

$$\mu_k = \frac{\sum_{n=1}^N \gamma(z_{nk}) \mathbf{x}_n}{\sum_{n=1}^N \gamma(z_{nk})}$$

$$\Sigma_k = \frac{\sum_{n=1}^N \gamma(z_{nk}) (\mathbf{x}_n - \mu_k)(\mathbf{x}_n - \mu_k)^T}{\sum_{n=1}^N \gamma(z_{nk})}$$

$$\pi_k = \frac{1}{N} \sum_{n=1}^N \gamma(z_{nk})$$

4. Evaluate log likelihood

$$\ln p(\mathbf{X} | \mu, \Sigma, \pi) = \sum_{n=1}^N \ln \left\{ \sum_{k=1}^K \pi_k \mathbf{N}(\mathbf{x}_n | \mu_k, \Sigma_k) \right\}$$

If there is no convergence, return to step 2.

Relationship to K-means

- K-means makes **hard** decisions.
 - Each data point gets assigned to a single cluster.
- GMM/EM makes **soft** decisions.
 - Each data point can yield a posterior $p(z|x)$
- K-means is a special case of EM.

Quick Knowledge Check

- **Which algorithm is used to fit the parameters of a GMM?** A) Expectation-Maximization (EM) B) Gradient Descent
- **During the E-step of EM, what do we compute?** A) The updated parameter values (μ_k, σ_k, π_k) B) The responsibilities $P(z_k | x)$
- **During the M-step of EM, what do we compute?** A) The responsibilities $P(z_k | x)$ B) The updated parameter values (μ_k, σ_k, π_k)
- **What does the mixture weight π_k represent?** A) The proportion of data points belonging to cluster k B) The variance of cluster k
- **If two Gaussian components overlap heavily, what happens to responsibilities?** A) They become close to equal for points in the overlapping region. B) They remain sharply separated

General form of EM

- Given a joint distribution over observed and latent variables: $p(X, Z|\theta)$
- Want to maximize: $p(X|\theta)$

1. Initialize parameters θ^{old}

2. E Step: Evaluate: $p(Z|X, \theta^{old})$

3. M-Step: Re-estimate parameters (based on expectation of complete-data log likelihood)

$$\theta^{new} = \operatorname{argmax}_{\theta} \sum_Z p(Z_k|X, \theta^{old}) \ln p(X, z_k|\theta) = \operatorname{argmax}_{\theta} \operatorname{Exp}[\log(p(x, z_k|\theta))]$$

1. Check for convergence of params or likelihood

The idea behind EM

Our log likelihood that we need to maximize is

$$\begin{aligned}\max l(\theta|x) &= \log p(x|\theta) = \log \sum_{k=0}^K \pi_k N(x|\mu_k, \Sigma_k) \\ &= \log \sum_{k=0}^K p(z_k|\theta) * p(x|z_k, \theta) \\ &= \log \sum_k p(x, z_k|\theta)\end{aligned}$$

Log of sum of gaussians is difficult to optimize

Let us introduce a helper distribution $q(z|x)$ and use Jensen inequality to help with this optimization (break log-sum)

Jensen's inequality $E [\log (g(x))] \leq \log E[g(x)]$

$$\ell(\theta | x) = \log p(x | \theta)$$

$$= \log \sum_k p(x, z_k | \theta)$$

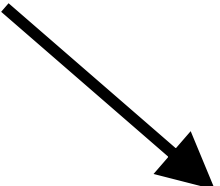
$$= \log \sum_k q(z_k | x) \frac{p(x, z_k | \theta)}{q(z_k | x)}$$

Will lead to maximize this

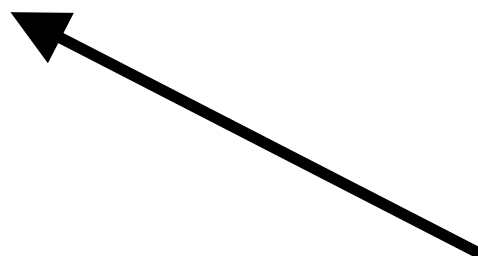
$$\geq \sum_k q(z_k | x) \log \frac{p(x, z_k | \theta)}{q(z_k | x)}$$

Maximizing this

Upper Bound


$$= \log \sum_k q(z_k | x) \frac{p(x, z_k | \theta)}{q(z_k | x)}$$

$$\geq \sum_k q(z_k | x) \log \frac{p(x, z_k | \theta)}{q(z_k | x)}$$



Lower Bound

In the M step of EM algorithm, we maximize this lower bound since it is easy to maximize to get new set of parameters

What happens when our helper distribution function
 $q(z|x) = p(z|x, \theta)$

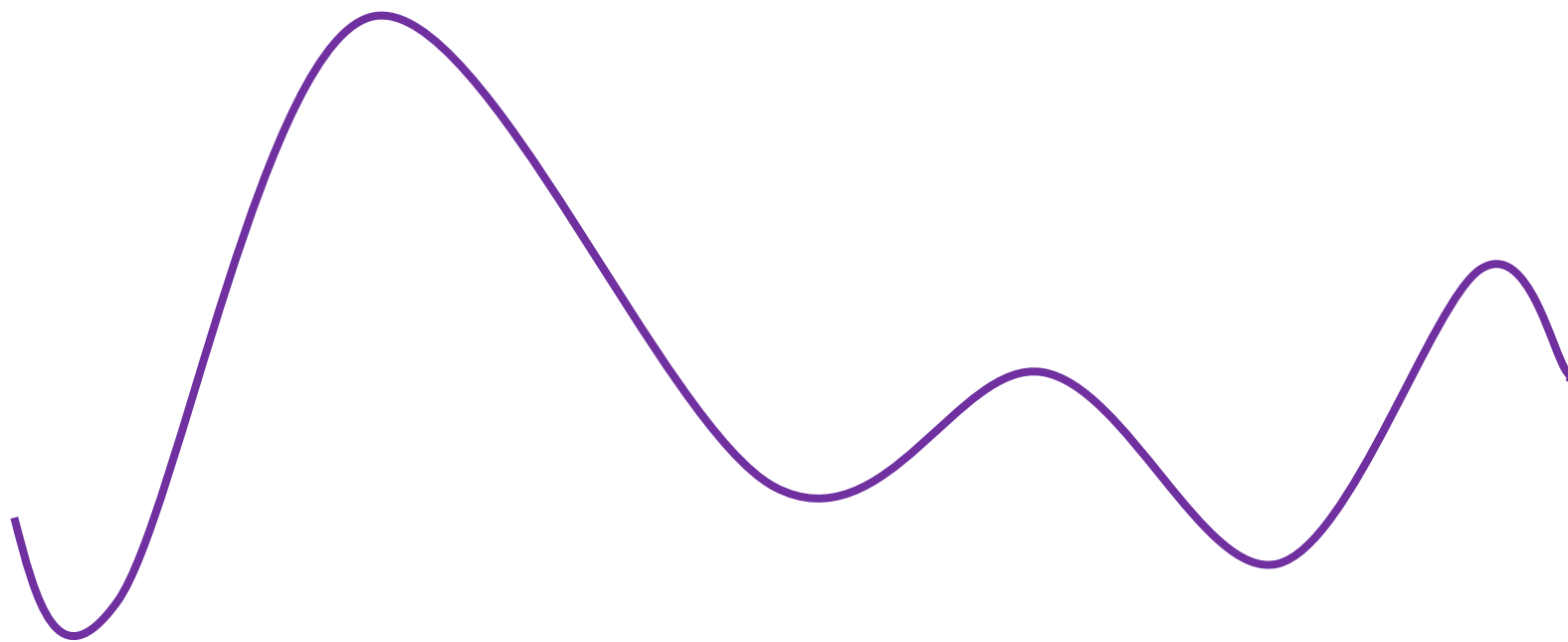
$$l(\theta|x) = \log p(x|\theta) = \log \sum_z p(x, z|\theta) \geq \sum_z q(z|x) \log \frac{p(x, z|\theta)}{q(z|x)}$$

We get equality of upper bound and lower bound. **This is called tight bound and what happens in E step of the algorithm**

Upper Bound

$$l(\theta|x) = \log p(x|\theta) = \log \sum_k p(x, z_k|\theta) \geq \sum_k q(z_k|x) \log \frac{p(x, z_k|\theta)}{q(z_k|x)}$$

$$\begin{aligned} \log \left(\sum_k p(x, z_k|\theta) \right) &= \log \left(\sum_z p(x|\theta, z_k) * p(z_k|\theta) \right) \\ &= \log(N(x|\mu_0, \Sigma_0) * \pi_0 + \dots + N(x|\mu_k, \Sigma_k) * \pi_0) \end{aligned}$$

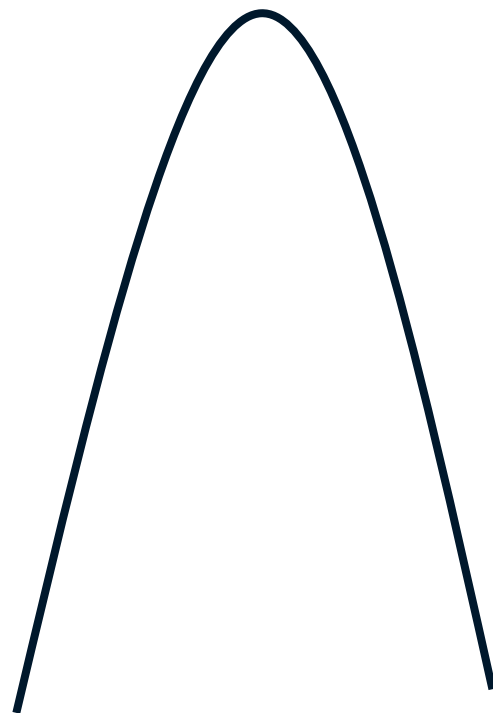


Lower Bound

$$l(\theta|x) = \log p(x|\theta) = \log \sum_k p(x, z_k|\theta) \geq \sum_k q(z_k|x) \log \frac{p(x, z_k|\theta)}{q(z_k|x)}$$

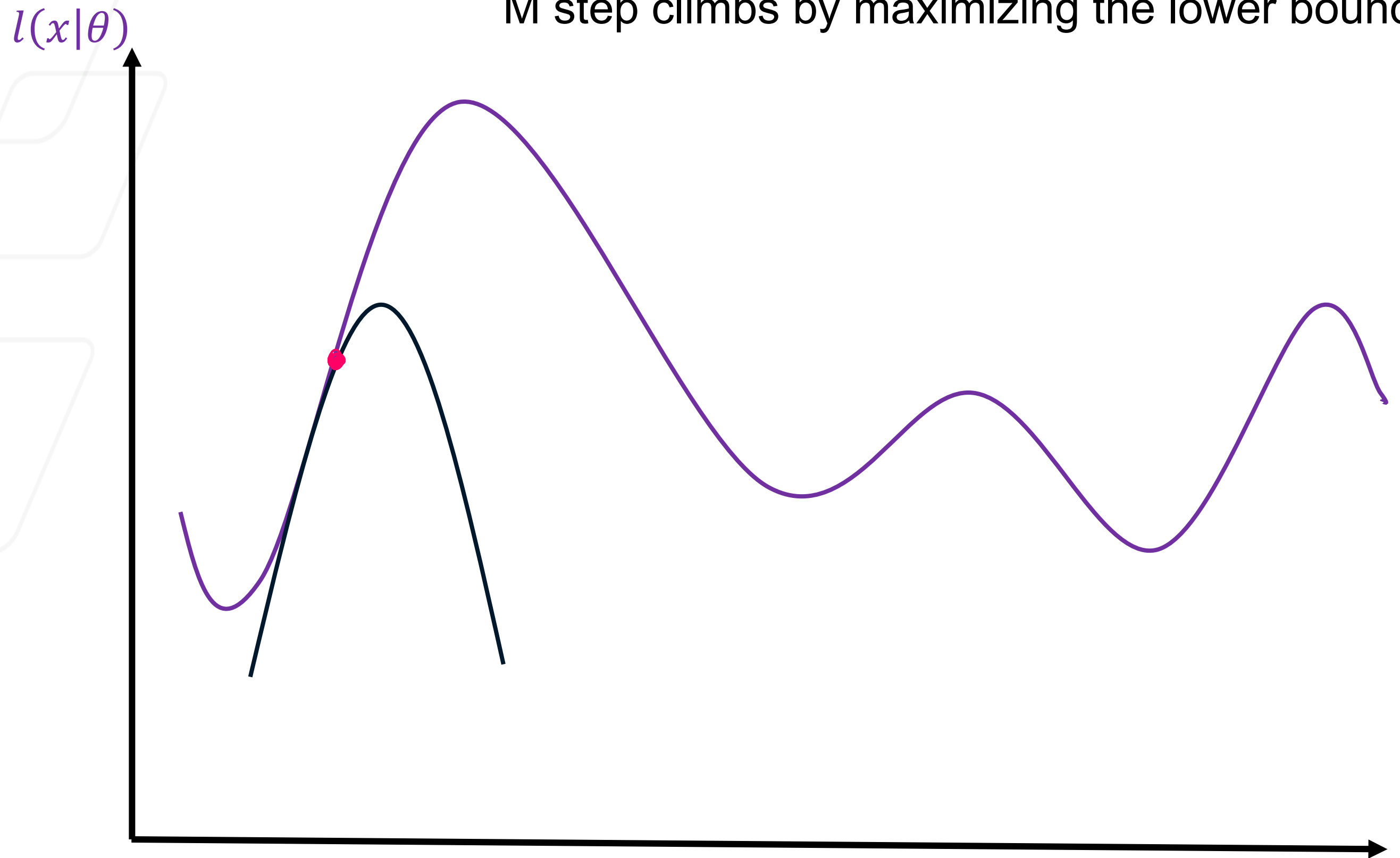
$q(z_k|x) = C_k \Rightarrow$ It is given to us

$$\sum_k q(z_k|x) \log \frac{p(x, z_k|\theta)}{q(z_k|x)} =$$
$$C_0 \log \left(\frac{1}{C_0} * N(x|\mu_0, \Sigma_0) * \pi_0 \right) + \dots + C_k \log \left(\frac{1}{C_k} * N(x|\mu_k, \Sigma_k) * \pi_k \right)$$



E step makes the bound tight

M step climbs by maximizing the lower bound



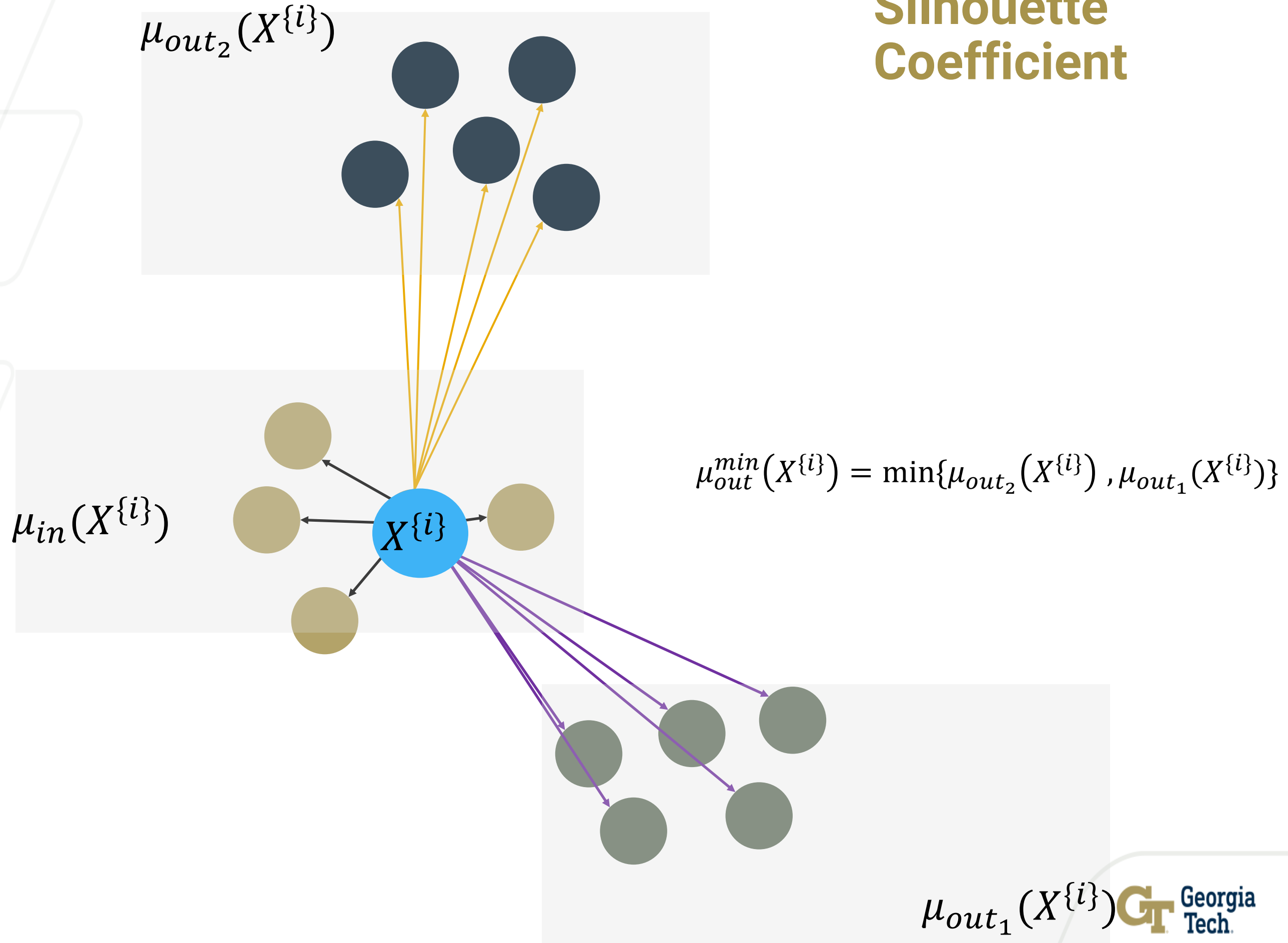
Summary

- MLE is hard because of **log-sum-exp** in the mixture.
- We **insert** q and use **Jensen** inequality to create a **lower bound**.
- This lower bound is sum of log and easy to optimize
- **E-step** sets $q = p(z' | x, \theta^{old}) \rightarrow$ the **bound becomes tight** at θ^{old}
- **M-step maximizes the lower bound**, so the **likelihood cannot decrease**.
- This iterative process continues until we get convergence (no more change of log likelihood – local maxima)
- Updates are just **weighted MLE** using responsibilities.

Take-Home Messages

- The generative process of Gaussian Mixture Model
- Inferring cluster membership based on a learned GMM
- The general idea of Expectation-Maximization
- Expectation-Maximization for GMM

Silhouette Coefficient



Silhouette Coefficient

Define the silhouette coefficient of a point $\mathbf{x}_i$ as

$$s_i = \frac{\mu_{out}^{\min}(\mathbf{x}_i) - \mu_{in}(\mathbf{x}_i)}{\max\{\mu_{out}^{\min}(\mathbf{x}_i), \mu_{in}(\mathbf{x}_i)\}}$$

where $\mu_{in}(\mathbf{x}_i)$ is the mean distance from $\mathbf{x}_i$ to points in its own cluster $\hat{y}_i$:

$$\mu_{in}(\mathbf{x}_i) = \frac{\sum_{\mathbf{x}_j \in C_{\hat{y}_i}, j \neq i} \delta(\mathbf{x}_i, \mathbf{x}_j)}{n_{\hat{y}_i} - 1}$$

and $\mu_{out}^{\min}(\mathbf{x}_i)$ is the mean of the distances from $\mathbf{x}_i$ to points in the closest cluster:

$$\mu_{out}^{\min}(\mathbf{x}_i) = \min_{j \neq \hat{y}_i} \left\{ \frac{\sum_{\mathbf{y} \in C_j} \delta(\mathbf{x}_i, \mathbf{y})}{n_j} \right\}$$

The Silhouette Coefficient for clustering C: $SC = \frac{1}{n} \sum_{i=1}^n s_i.$

SC close to 1 implies a good clustering (Points are close to their own clusters but far from other clusters)